

**SYLLABUS
ECONOMICS FOR ENGINEERS
(HS-301)**

Applicable from Batch Admitted in Academic Session 2021-22 onwards

Marking Scheme:

1. Teachers Continuous Evaluation: 25 marks
2. Term end Theory Examinations: 75 marks

Instructions for paper setter:

1. There should be 9 questions in the term end examinations question paper.
2. The first (Ist) question should be compulsory and cover the entire syllabus. This question should be objective, single line answers or short answer type question of total 15 marks.
3. Apart from question 1 which is compulsory, rest of the paper shall consist of 4 units as per the syllabus. Every unit shall have two questions covering the corresponding unit of the syllabus. However, the student shall be asked to attempt only one of the two questions in the unit. Individual questions may contain upto 5 sub-parts / sub-questions. Each Unit shall have a marks weightage of 15.
4. The questions are to be framed keeping in view the learning outcomes of the course / paper. The standard / level of the questions to be asked should be at the level of the prescribed textbook.
5. The requirement of (scientific) calculators / log-tables / data - tables may be specified if required.

UNIT I

Introduction: Economics Definition, Basic economic problems, Resource constraints and welfare maximization. Micro and Macro economics. Production Possibility Curve. Circular flow of economic activities.

Basics of Demand, Supply and Equilibrium: Demand side and supply side of the market. Factors affecting demand & supply. Elasticity of demand & supply - price, income and cross-price elasticity. Market equilibrium price.

UNIT II

Theory of Consumer Choice: Theory of Utility and consumer's equilibrium. Indifference Curve analysis, Budget Constraints, Consumer Equilibrium.
Demand forecasting: Regression Technique, Time-series, Smoothing Techniques: Exponential, Moving Averages Method

UNIT III

Cost Theory and Analysis: Nature and types of cost, Cost functions- short run and long run, Economies and diseconomies of scale
Market Structure: Market structure and degree of competition, Perfect competition, Monopoly, Monopolistic competition, Oligopoly

UNIT IV

National Income Accounting: Overview of Macroeconomics, Basic concepts of National Income Accounting
Macro Economics Issues: Introduction to Business Cycle, Inflation-causes, consequences and remedies: Monetary and Fiscal policy.

MODEL PAPER-1
FIFTH SEMESTER [B.TECH]
ECONOMICS FOR ENGINEERS [HS-301]

Q.1. Explain the following in short:

Q.1. (a) Define the economics.

Ans. Economics is a social science that focuses on the production, distribution, and consumption of goods and services, and analyses the choices that individuals, businesses, governments, and nations make to allocate resources.

Economics can be defined in a few different ways. It's the study of scarcity, the study of how people use resources and respond to incentives, or the study of decision-making. It often involves topics like wealth and finance, but it's not all about money. Economics is a broad discipline that helps us understand historical trends, interpret today's headlines, and make predictions about the coming years.

Types of Economics

There are two main types of economics: microeconomics and macroeconomics.

(i) Microeconomics

The study of microeconomics focuses on the choices of individuals and businesses, and macroeconomics concentrates on the behaviour of the economy as a whole, on an aggregate level.

Microeconomics studies individuals' and businesses' economic decisions and how those decisions affect resources. For instance, what happens if a running gear store raises the price of its shoes? Will consumers still buy them, and if so, how many? Economists who work with businesses use microeconomics to help forecast outcomes and advise economic decisions.

(ii) Macroeconomics

Macroeconomics is the study of the economy as a whole. In this kind of economics, economists look at large-scale factors like inflation, unemployment, and national income. Their goal is to see how the economy is performing as a whole, why it's performing the way it is, and how it can improve.

Q.1. (b) Time Series

Ans. Time series refers to a sequence of data points collected and recorded over a specific time interval. These data points are typically ordered chronologically, making time a crucial factor in the analysis and interpretation of the data. Time series data can come from various sources, such as financial markets, weather measurements, economic indicators, stock prices, sensor readings, and more.

Key characteristics of time series data include:

1. **Temporal Order:** Data points are collected in a sequential order, where each point corresponds to a specific time period.
2. **Time Dependency:** The data points are often interdependent, meaning that the value of a data point at a given time is influenced by previous data points.
3. **Seasonality:** Many time series exhibit regular patterns or cycles, such as daily, weekly, monthly, or yearly fluctuations. These patterns are known as seasonality.
4. **Trends:** Time series data can exhibit trends, which are long-term changes or shifts in the data over time. Trends can be upward (increasing values), downward (decreasing values), or flat (no significant change).
5. **Noise:** Time series data often contain random fluctuations or noise, which can obscure underlying patterns. Analysing time series data involves several tasks, including:

6. Data Exploration: Understanding the basic characteristics of the data, identifying trends, seasonality, and outliers.

7. Forecasting: Predicting future values based on historical data. Techniques like moving averages, exponential smoothing, and advanced methods like ARIMA (AutoRegressive Integrated Moving Average) and machine learning models are used for forecasting.

8. Seasonal Decomposition: Separating a time series into its underlying components, such as trend, seasonality, and residual (noise), to better understand its behavior.

9. Anomaly Detection: Identifying abnormal or unexpected patterns in the data that deviate from the norm.

10. Causal Analysis: Exploring relationships between time series data and external factors to understand how one might influence the other.

11. Smoothing: Applying techniques to reduce noise and emphasize underlying patterns.

12. Feature Engineering: Creating relevant features from the time series data for machine learning algorithms.

13. Modeling: Building mathematical, statistical, or machine learning models to make predictions, detect patterns, or perform other analysis.

Q.1. (c) Opportunity cost.

Ans. Opportunity cost is a fundamental concept in economics that refers to the value of the next best alternative that must be foregone when a decision is made to allocate resources, such as time, money, or effort, to a particular choice. In other words, it's the cost of choosing one option over another.

When you make a decision, there are often multiple options available to you, each with its own benefits and drawbacks. The opportunity cost is what you give up in terms of potential benefits from the next best alternative that you didn't choose. This concept highlights the trade-offs involved in decision-making and helps individuals and businesses make more informed choices.

Opportunity costs represent the potential benefits that an individual, investor, or business misses out on when choosing one alternative over another. Because opportunity costs are unseen by definition, they can be easily overlooked. Understanding the potential missed opportunities when a business or individual chooses one investment over another allows for better decision making.

Opportunity cost cannot always be fully quantified at the time when a decision is made. Instead, the person making the decision can only roughly estimate the outcomes of various alternatives, which means imperfect knowledge can lead to an opportunity cost that will only become obvious in retrospect. This is a particular concern when there is a high variability of return. To return to the first example, the foregone investment at 7% might have a high variability of return, and so might not generate the full 7% return over the life of the investment.

While opportunity cost is not an exact measure, one way to quantify it is to estimate the potential future value that you opted not to receive and compare it with the value of the choice you made instead.

Q.1. (d) Market Demand.

Ans. Market demand is the total quantity of a product or service that consumers are willing and able to purchase at a given price and within a specific market. It represents the collective desire and purchasing power of potential customers. Market demand is influenced by factors such as consumer preferences, income levels, population demographics, and overall economic conditions.

Total market demand can fluctuate over time—in most cases, it does. This could be due to a variety of factors, some seasonal and predictable, others more out of our control, like a natural disaster or even a pandemic. Sometimes the entire demand curve shifts.

Market demand is the aggregate of the individual demands for a commodity from purchasers in the marketplace. If more purchasers enter the marketplace and they have the capability to pay for commodities on sale, then the market demand at each cost price degree will increase.

In the previous section, we studied the preference of individual customers and derived the demand curve for it. However, there are many customers for a commodity in the marketplace. It is important to find out the market demand for the commodity.

The market demand for a commodity at a particular cost price is the total demand of all the customers taken together. The market demand for a commodity can be derived from the individual demand curves. It can also be derived from the individual demand curves graphical depiction by summing up the individual demand curves.

When more people want a specific type of product, this is an increase in market demand and prices typically go up—more people want it and more people are willing to pay for it. But when market demand decreases, prices typically follow suit. It gets more complex than that, but we'll get into it later.

Q.1. (e) Macro Economics.

Ans. Macroeconomics is a branch of economics that depicts a substantial picture. It scrutinises itself with the economy at a massive scale and several issues of an economy are considered. The issues confronted by an economy and the headway that it makes are measured and apprehended as a part and parcel of macroeconomics. When one speaks of the issues that an economy confronts, inflation, unemployment, increasing tax burden, etc., are all contemplated. This makes it apparent that macroeconomics focuses on large numbers.

It studies the association between various countries regarding how the policies of one nation have an upshot on the other. It circumscribes within its scope, analysing the success and failure of government strategies.

Concepts covered under macroeconomics

A capitalist nation

A capitalist country is distinguished by sub-urbanised and voluntary conclusions for economic planning instead of the consolidated political practices. There are a few aspects of a capitalist financial structure (Economy) mentioned that would provide a better intuition into the concept. The attributes of a capitalist nation are as follows:

1. Liberty of customers to pick between goods and services.
2. The privilege of individuals to set up a business to supply goods and services.
3. There is a finite interference of the government.
4. Market forces regulate the distribution of goods.

Investment expenditure

As the name says it all, it is the money consumed towards charges to create investments. In other words, it is the money that the family circle (households) and enterprises spend on capital goods. It plays a decisive role in macroeconomic pursuit for business cycles and economic enhancement in the long run.

In short, the investment expenditure is proficient of creating additional income and fosters employment in a nation.

The following are the types of investments:

1. Autonomous investment
2. Financial investment
3. Real investment
4. Gross investment
5. Net investment

Revenue: Revenue is the total income of an entity through sale of goods and proffering its services to the customers. Revenue can be operating or non-operating. The significance of revenue and its acknowledgements is better comprehended if we are well aware of the aspects that are contemplated while deciding the GDP.

Q.1. (f) Profit Maximization.

Ans. Profit maximisation is a process business firms undergo to ensure the best output and price levels are achieved in order to maximise its returns.

Influential factors such as sale price, production cost and output levels are adjusted by the firm as a way of realising its profit goals.

In business, profit maximisation is a good thing, but it can be a bad thing for the client if, for example, lower-quality materials and labour are used or if the business decides to raise the prices for executing projects, all in pursuit of profit maximisation.

Profit maximization is the process by which a business or an organization aims to generate the highest possible amount of profit within a given time period, typically considering its costs, revenues, and various economic factors. In a simplified economic context, profit can be calculated by subtracting total costs from total revenues. The goal of profit maximization is to find the optimal balance between these factors in order to achieve the highest level of profit.

There are different approaches to pursuing profit maximization, and they can vary based on the specific goals and circumstances of a business:

1. Cost Minimization- One approach to profit maximization is to focus on minimizing costs while maintaining a certain level of quality and production. By reducing costs, a business can potentially increase its profit margins.

2. Revenue Maximization- Another approach is to focus on increasing total revenue by optimizing pricing, sales volume, and market share. This might involve strategies like product differentiation, advertising, or expanding into new markets.

3. Economies of Scale- Some businesses aim to maximize profit by achieving economies of scale, where the cost per unit of production decreases as production volume increases. This allows for higher profit margins on each unit produced.

4. Marginal Analysis- Businesses can use marginal analysis to determine the optimal level of production or output at which marginal revenue (the additional revenue from producing one more unit) equals marginal cost (the additional cost of producing one more unit). At this point, profit is maximized.

5. Long term and short term perspective- Profit maximization can be approached from both short-term and long-term perspectives. In the short term, businesses might make decisions that lead to immediate profit increases, while in the long term, they might invest in research and development, employee training, and other strategies that could lead to sustained profit growth.

It's worth noting that while profit maximization is a common goal for businesses, it's not the only factor that organizations consider. Other objectives, such as market share growth, customer satisfaction, sustainability, and corporate social responsibility, can also play important roles in shaping business strategies and decision-making.

Q.1. (g) Monopoly Market.

Ans. A monopoly market is a type of market structure where a single seller or producer dominates the entire market for a particular product or service. In a monopoly, there is no close competition, and the monopolistic firm becomes the sole provider of that product or service. This gives the monopolistic entity significant control over the market and allows them to set prices and output levels without being constrained by competition.

Key characteristics of a monopoly market include:

(i) **Single Seller:** There is only one firm operating in the market. This firm is the sole producer and supplier of the product or service.

(ii) **High Barriers to Entry:** Monopolies often arise due to significant barriers to entry that prevents other firms from entering the market and competing. Barriers to entry can include high startup costs, exclusive access to resources, legal barriers, and economies of scale that give the existing firm a cost advantage.

(iii) **Price Maker:** Since there is no competition, the monopolistic firm has the power to set prices according to its own discretion. It can choose to maximize profits by setting prices higher than would be possible in a competitive market.

(iv) **Limited Consumer Choice:** Consumers have limited or no choice but to buy from the monopolist, which can lead to reduced variety and potentially lower quality.

(v) **Market Power:** Monopolies have significant market power, allowing them to influence the overall market demand, supply, and price.

(vi) **Potential for Inefficiency:** Monopolies may lack the competitive pressure to innovate, improve efficiency, or provide optimal levels of production and quality, leading to potential inefficiencies.

(vii) **Regulation:** Due to concerns about abuse of market power and potential negative effects on consumers, governments often regulate monopolies to ensure fair pricing, quality, and access.

(viii) **Natural Monopoly vs. Legal Monopoly:** Monopolies can arise for different reasons. Natural monopolies occur when it's more efficient to have a single provider due to economies of scale (e.g., public utilities like water and electricity). Legal monopolies are established through government-granted licenses or patents that provide exclusive rights to produce or sell a certain product or service.

Q.2. What do you understand by Production Possibility Curve. How it defines the economic problems. Explain with the help of diagram.

Ans. The production possibility curve represents graphically alternative production possibilities open to an economy.

The productive resources of the community can be used for the production of various alternative goods.

But since they are scarce, a choice has to be made between the alternative goods that can be produced. In other words, the economy has to choose which goods to produce and in what quantities. If it is decided to produce more of certain goods, the production of certain other goods has to be curtailed.

Let us suppose that the economy can produce two commodities, cotton and wheat. We suppose that the productive resources are being fully utilized and there is no change in technology. The following table gives the various production possibilities.

Production Possibilities	Cotton (in 000 quintals)	Wheat (in 000 quintals)
A	0	15
B	1	14
C	2	12
D	3	9
E	4	5
F	5	0

If all available resources are employed for the production of wheat, 15,000 quintals of it can be produced. If, on the other hand, all available resources are utilized for the production of cotton, 5000 quintals are produced. These are the two extremes represented by A and F and in between them are the situations represented by B, C, D and E. At B, the economy can produce 14,000 quintals of wheat and 1000 quintals of cotton.

At C the production possibilities are 12,000 quintals of wheat and 2000 quintals of cotton, as we move from A to F, we give up some units of wheat for some units of cotton. For instance, moving from A to B, we sacrifice 1000 quintals of wheat to produce 1000 quintals of cotton, and so on. As we move from A to F, we sacrifice increasing amounts of cotton.

This means that, in a full-employment economy, more and more of one good can be obtained only by reducing the production of another good. This is due to the basic fact that the economy's resources are limited.

The following diagram (1) illustrates the production possibilities set out in the above table.

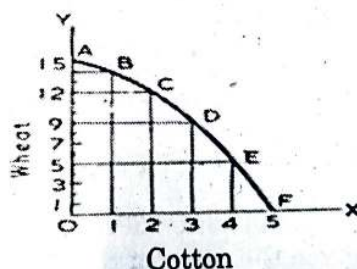


Fig. 1 Production Possibility Curve

In this diagram AF is the production possibility curve, also called or the production possibility frontier, which shows the various combinations of the two goods which the economy can produce with a given amount of resources. The production possibility curve is also called transformation curve, because when we move from one position to another, we are really transforming one good into another by shifting resources from one use to another.

It is to be remembered that all the points representing the various reduction possibilities must lie on the production possibility curve AF and not inside or outside of it. For example, the combined output of the two goods can neither be at U nor H. (See Fig. 2) This is so because at U the economy will be under-employing its resources and H is beyond the resources available.

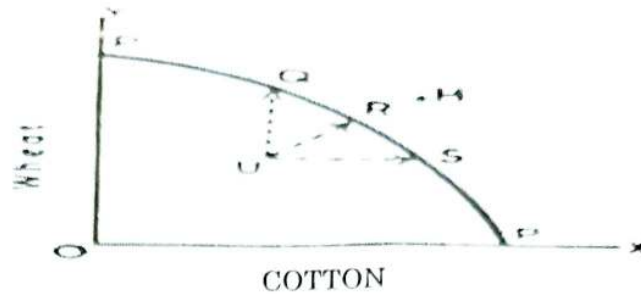


Fig. 2 Problem of Under-Utilisation of Resources

The central problem of an economy, often referred to as the fundamental economic problem, is how to effectively allocate scarce resources among competing uses or wants. This problem arises due to the limited availability of resources (such as labor, land, capital, and technology) in relation to unlimited human wants and needs. The production possibility curve (PPC), also known as the production possibility frontier (PPF), is closely related to this central economic problem as it helps illustrate the trade-offs and choices that arise when allocating these scarce resources.

The PPC graphically represents the combinations of two goods or services that an economy can produce using its available resources and technology. The curve demonstrates the trade-off between producing one good over the other, given the limited resources. The PPC embodies the key economic problem in several ways:

1. Scarcity: The PPC reflects the limited nature of resources. It shows that an economy cannot produce unlimited quantities of both goods simultaneously. Choices have to be made regarding what to produce.

2. Choice: The PPC illustrates the necessity of making choices due to scarcity. An economy must decide which combination of goods to produce along the curve. Producing more of one good inevitably means producing less of another.

3. Opportunity Cost: The PPC highlights the concept of opportunity cost—the value of the next best alternative foregone when a decision is made. Moving along the PPC involves trading one good for another, and the slope of the curve represents the opportunity cost of producing more of one good in terms of the quantity of the other good given up.

4. Efficiency: Points along the PPC represent efficient resource allocation, where an economy is using all available resources to produce a combination of goods. Points inside the curve represent underutilization of resources, while points outside the curve are currently unattainable given the existing resources and technology.

5. Dynamic Nature: The PPC can shift outward over time due to factors such as technological advancements, increased resources, or improvements in resource utilization. This expansion of the curve represents economic growth.

In essence, the central problem of an economy is how to make optimal choices about resource allocation given the trade-offs between different goods and services. The PPC serves as a visual representation of these choices and trade-offs, helping economists and policymakers understand the implications of resource allocation decisions and the potential for economic growth.

Q.3. What is demand forecasting? Explain the various methods that are used while computing the demand forecasting.

Ans. Demand forecasting is a process used by businesses and organizations to predict future customer demand for their products or services. It involves estimating the quantity of goods or services that customers are likely to purchase within a specific

time frame. Accurate demand forecasting is crucial for effective inventory management, production planning, resource allocation, and overall business strategy development.

Various methods are employed to compute demand forecasting, and they can be broadly categorized into qualitative and quantitative methods.

(i) Qualitative Methods: Qualitative methods rely on subjective judgment, expert opinions, and non-quantitative factors to predict demand. These methods are often used when historical data is limited or when there are significant changes in market conditions. Some common qualitative methods include:

(a) Market Research: Gathering data from surveys, focus groups, and interviews to understand customer preferences, behaviors, and expectations.

(b) Expert Opinion: Consulting industry experts, salespeople, and managers to gather insights on market trends and potential demand.

(c) Delphi Method: A structured approach that involves multiple rounds of anonymous surveys with experts to reach a consensus on future demand.

(d) Scenario Analysis: Developing various scenarios based on different assumptions about market conditions, regulatory changes, economic factors, and technological advancements.

(e) Consumer Surveys: Collecting information directly from customers through surveys or questionnaires to gauge their future buying intentions.

(ii) Quantitative Methods: Quantitative methods rely on historical data and statistical techniques to forecast future demand. These methods are more data-driven and suitable when historical data is available and relevant. Some common quantitative methods include:

(a) Time Series Analysis: Analyzing historical sales data over time to identify patterns, trends, and seasonality. Techniques like moving averages, exponential smoothing, and ARIMA (AutoRegressive Integrated Moving Average) models are used.

(b) Causal Methods: Examining the relationship between demand and specific influencing factors, such as price, advertising expenditure, and economic indicators. Linear regression and multiple regression models are commonly used.

(c) Trend Analysis: Identifying long-term trends in historical data and extrapolating them to predict future demand.

(d) Seasonal Decomposition: Separating historical data into trend, seasonal, and irregular components to better understand and forecast demand patterns.

(e) Machine Learning and Artificial Intelligence: Using advanced techniques like neural networks, support vector machines, and random forests to analyze large datasets and predict future demand based on patterns and correlations.

(f) Econometric Models: Applying economic theories and models to forecast demand by considering factors like income levels, population growth, and consumer behaviour.

(g) Simulation Models: Creating computer models that simulate the behavior of consumers, competitors, and other market variables to predict future demand under various scenarios.

Choosing the appropriate method depends on factors such as data availability, the nature of the product or service, the level of accuracy required, and the specific market conditions. In practice, a combination of qualitative and quantitative methods is often used to provide a more comprehensive and accurate demand forecast.

Q.4. Explain the characteristics of Long term production function.

Ans. The stages of production function, also known as the stages of production,

refer to the different phases through which the production process of a good or service progresses as the quantity of inputs used in production increases. These stages are important for understanding how changes in input levels affect output and productivity. The stages are characterized by changes in the marginal product of inputs and provide insights into the efficiency of resource utilization. There are three main stages of production: the increasing returns to scale stage, the diminishing returns to scale stage, and the negative returns to scale stage.

Stage 1: Increasing Returns to Scale (Increasing Marginal Returns):

In this stage, as additional units of input (e.g., labor) are added while keeping other inputs constant, the total output increases at an increasing rate. This means that the marginal product of the added input is positive and rising.

Characteristics:

1. High levels of specialization and division of labour.
2. Efficient utilization of resources due to specialization.
3. Economies of scale leading to reduced average costs.

Stage 2: Diminishing Returns to Scale (Diminishing Marginal Returns):

In this stage, as more units of input are added, the total output continues to increase but at a decreasing rate. The marginal product of the added input starts to decline.

Characteristics:

1. Resources are still being effectively utilized but at a slower pace.
2. Marginal product starts to decline, leading to less efficient resource utilization.
3. Output continues to grow, but the rate of growth slows down.

Stage 3: Negative Returns to Scale (Negative Marginal Returns):

In this stage, adding more units of input causes total output to decrease. The marginal product of the added input becomes negative, and resources are being overutilized or mismanaged.

Characteristics:

1. Resources are being stretched beyond their optimal capacity.
2. Overcrowding and inefficiencies lead to reduced output.
3. Marginal product becomes negative, and average costs rise.

It's important to note that while these stages provide a general framework, the precise points at which each stage begins and ends can vary depending on the specific circumstances of production, the nature of the inputs, and other factors. The concept of stages of production helps businesses and economists understand the relationship between input levels and output, aiding in making informed decisions about resource allocation, production planning, and cost management.

Overall, the goal for producers is to operate within the stage where marginal product is positive and decreasing, as this represents the region of optimal resource utilization and efficiency.

Q.5. Explain the role of demand and supply in price determination.

Ans. Demand and supply are fundamental concepts in economics that play a crucial role in determining the prices of goods and services in a market economy. They describe the relationship between the quantity of a product that consumers are willing to buy (demand) and the quantity that producers are willing to produce and sell (supply) at

various price levels. The interaction between demand and supply ultimately determines the equilibrium price and quantity in a market. Demand and supply are fundamental concepts in economics that play a crucial role in determining the prices of goods and services in a market economy. They describe the relationship between the quantity of a product that consumers are willing to buy (demand) and the quantity that producers are willing to produce and sell (supply) at various price levels. The interaction between demand and supply ultimately determines the equilibrium price and quantity in a market.

(a) Demand: Demand refers to the quantity of a good or service that consumers are willing and able to purchase at various price levels within a given period of time. Several factors influence demand:

1. Price of the Product: As the price of a product decreases, consumers generally tend to buy more of it, and as the price increases, they buy less. This relationship is described by the law of demand.

2. Income: If people's income increases, they might be more willing to buy certain goods and services even at higher prices. Conversely, if income decreases, they might cut back on certain purchases.

3. Price of Related Goods: If the price of a substitute product (a product that can be used in place of another) increases, demand for the original product might increase. On the other hand, if the price of a complementary product (a product often used together with another) increases, demand for the original product might decrease.

4. Consumer Preferences: Changes in consumer preferences or trends can significantly impact the demand for a product. For example, if a product becomes more popular, its demand might increase.

5. Population and Demographics: The size and composition of the population also influence demand. A larger population generally leads to higher demand for goods and services.

(b) Supply: Supply refers to the quantity of a good or service that producers are willing and able to offer for sale at various price levels within a given period of time. Several factors influence supply:

1. Price of the Product: If the price of a product increases, producers are more likely to produce and supply more of it, as it becomes more profitable. This relationship is described by the law of supply.

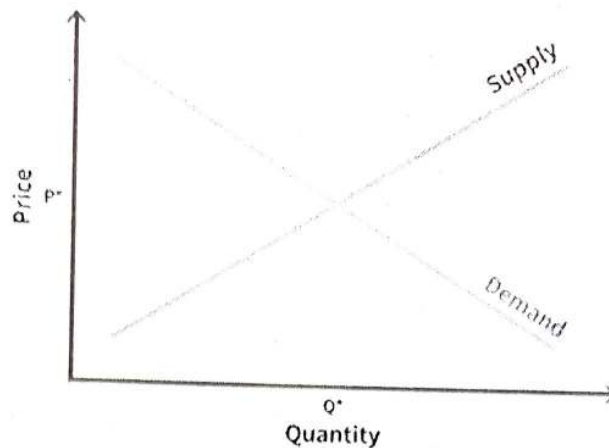
2. Production Costs: The costs involved in producing a product, including factors such as labor, raw materials, and technology, impact the willingness of producers to supply the product. If production costs rise, supply might decrease.

3. Technology and Innovation: Advances in technology can lead to increased production efficiency, which can influence the level of supply.

4. Number of Producers: The number of firms or producers in a market can affect supply. More producers can lead to greater supply competition.

5. Government Policies: Government regulations, taxes, subsidies, and other policies can impact the cost of production and thus influence supply.

Price Determination: The equilibrium price in a market is determined by the point at which the demand and supply curves intersect. This equilibrium price is also known as the market-clearing price. At this price, the quantity demanded equals the quantity supplied, resulting in a balance between what consumers want to buy and what producers want to sell.



If the price is set above the equilibrium level (a price floor), there will be a surplus of the product as producers are willing to supply more than consumers are willing to buy. Conversely, if the price is set below the equilibrium level (a price ceiling), there will be a shortage of the product as consumers demand more than producers are willing to supply.

In summary, demand and supply are essential forces that interact to determine the prices of goods and services in a market economy. Changes in either demand or supply can lead to shifts in the equilibrium price and quantity, influencing the overall dynamics of the market.

Q.6. Define the Business Cycle in detail. And also explain the reason for every stages of Business Cycle.

Ans. The business cycle, also known as the economic cycle or trade cycle, is a recurring pattern of fluctuations in economic activity characterized by alternating periods of expansion and contraction in an economy. It reflects the natural ups and downs in economic growth, employment, and production. The business cycle consists of four main phases: expansion, peak, contraction, and trough. Let's delve into each phase and the reasons behind them:

(a) Expansion:

During an expansion phase, economic activity grows, leading to increased production, employment, consumer spending, and investment.

Reasons for the expansion phase:

(i) Increased Consumer Confidence: Positive economic indicators, such as low unemployment and rising income levels, boost consumer confidence. Consumers feel more secure about their financial situations and are more willing to spend.

(ii) Business Investment: Improved economic conditions encourage businesses to invest in expansion, new projects, and technological advancements, leading to increased production capacity.

(iii) Credit Availability: Favorable economic conditions usually result in easier access to credit and loans, enabling consumers and businesses to spend more.

(iv) Fiscal and Monetary Policies: Government policies such as tax cuts, infrastructure spending, and central bank interventions to lower interest rates can stimulate economic growth.

(b) Peak: The peak marks the highest point of economic expansion, characterized by high levels of economic activity and growth.

Reasons for the peak phase:

(i) Resource Constraints: As the economy reaches full capacity, resources like

labor and raw materials become scarce. This can lead to inflationary pressures.

(ii) Rising Interest Rates: Strong economic performance can prompt central banks to raise interest rates to prevent excessive borrowing and spending, which can contribute to inflation.

(iii) Speculative Behavior: In periods of prolonged growth, investors and consumers might become overly optimistic and engage in speculative activities, contributing to asset bubbles.

(c) Contraction (Recession or Downturn):

The contraction phase, also known as a recession, is characterized by a decline in economic activity, leading to reduced production, employment, consumer spending, and business investment.

Reasons for the contraction phase:

(i) Demand Shocks: External events, such as a financial crisis, geopolitical tensions, or a sharp decline in consumer and business confidence, can lead to a sudden decrease in demand for goods and services.

(ii) Tightening Monetary Policy: Central banks might raise interest rates to combat inflation, but this can also slow down borrowing and spending, leading to reduced economic activity.

(iii) High Debt Levels: If consumers and businesses accumulate high levels of debt during the expansion phase, they might cut back on spending during tougher times to manage their financial obligations.

(d) Trough:

The trough is the lowest point of the business cycle, where economic activity is at its lowest before it starts to recover.

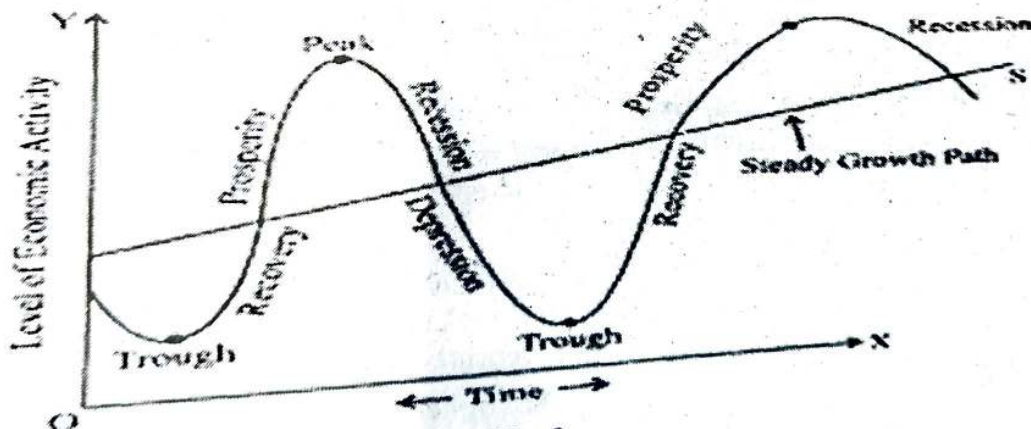


Fig. 3.

Reasons for the trough phase:

(i) Market Adjustment: As the economy experiences contraction, prices, wages, and asset values might decrease to adjust to lower demand and production levels.

(ii) Stabilizing Measures: Government intervention, such as stimulus packages, infrastructure projects, and lowered interest rates, can help stabilize the economy and support recovery.

(iii) Correction of Imbalances: During the trough, excesses accumulated during the expansion phase, like inflated asset prices or overproduction, are corrected.

It's important to note that the business cycle is a natural and inherent part of a market economy. The length and severity of each phase can vary widely depending

on economic factors, government policies, technological advancements, and external shocks. Economists and policymakers analyze these cycles to better understand economic trends, make informed decisions, and implement measures to mitigate the negative impacts of downturns and enhance the benefits of expansions.

Q.7. How Monetary policy is used at the time of Inflation and what are their effect on economic growth.

Ans. Monetary policy is a tool used by central banks to control the money supply, interest rates, and overall economic activity in a country. It is employed to achieve specific economic objectives, such as controlling inflation and promoting economic growth. When addressing inflation, central banks typically use contractionary monetary policy measures. Here's how monetary policy is used during inflation and its effects on economic growth:

(a) Monetary Policy During Inflation: Inflation occurs when the general price level of goods and services in an economy rises over time. To combat inflation and prevent it from spiraling out of control, central banks may implement contractionary monetary policy measures. The primary goal is to reduce the money supply, curb spending, and moderate the overall demand in the economy. The two main tools used for this purpose are:

(i) Increasing Interest Rates: Central banks can raise their policy interest rates, such as the federal funds rate in the United States, to make borrowing more expensive. This encourages consumers and businesses to reduce borrowing and spending, which in turn decreases aggregate demand.

(ii) Open Market Operations: Central banks can also sell government securities in open market operations, thereby reducing the money supply. When individuals and financial institutions buy these securities, they reduce their available funds for spending and investment, which helps dampen demand.

(b) Effects on Economic Growth: While contractionary monetary policy measures are effective in curbing inflation, they can also have effects on economic growth:

1. Slower Economic Growth: Higher interest rates make borrowing more expensive for consumers and businesses, leading to reduced spending on big-ticket items like homes and cars. This decrease in spending can slow down economic growth, as consumer spending is a significant driver of economic activity.

2. Reduced Business Investment: Higher interest rates can lead to increased borrowing costs for businesses looking to invest in expansion or new projects. This can deter investment and limit the growth potential of businesses.

3. Investment and Capital Flows: Higher interest rates might attract foreign capital looking for higher returns, potentially affecting exchange rates and capital flows. This can impact exports and imports, which can, in turn, influence economic growth.

4. Long-Term Benefits: If successful in reducing inflation, contractionary monetary policy can contribute to price stability, which is crucial for long-term economic planning and investment.

5. Economic Stabilization: While contractionary monetary policy might initially slow economic growth, it can help stabilize the economy by preventing runaway inflation and the negative consequences associated with it.

6. It's important to note that the effectiveness of monetary policy depends on various factors, including the transmission mechanism through which changes in monetary policy affect the broader economy, the responsiveness of households and businesses to

interest rate changes, and the overall economic context. Additionally, monetary policy works in conjunction with fiscal policy (government spending and taxation) to achieve macroeconomic goals.

Q.8. Write the characteristics of Macroeconomics and how it different with Micro economic. Explain

Ans. Macroeconomics and microeconomics are two branches of economics that study different aspects of the economy. They have distinct focuses and characteristics. Let's delve into the characteristics of macroeconomics and highlight the differences between macroeconomics and microeconomics:

Characteristics of Macroeconomics:

(a) Aggregates: Macroeconomics deals with the economy as a whole and focuses on aggregated variables such as total output (GDP), inflation, unemployment, and overall economic growth.

(b) National-Level Analysis: It analyzes the economic performance of a country or region as a single entity, considering factors that affect the entire economy.

(c) Global Impact: Macroeconomic factors can have global implications. Events like financial crises or changes in global trade relationships can affect economies around the world.

(d) Policy Implications: Macroeconomics is closely linked to policy-making. Governments and central banks use macroeconomic tools (monetary and fiscal policies) to manage economic growth, stability, and employment.

(e) Key Variables: Key variables in macroeconomics include Gross Domestic Product (GDP), inflation rate, unemployment rate, interest rates, and overall economic indicators.

Long-Term Trends: Macroeconomics often focuses on analyzing long-term trends and cycles in the economy, such as business cycles and economic growth over time.

(f) Systemic Risks: Macroeconomics examines systemic risks that can affect the stability of the financial system, such as banking crises and market crashes.

Differences between Macroeconomics and Microeconomics:

Scope: Macroeconomics deals with the economy as a whole and examines broad economic aggregates.

Microeconomics focuses on individual markets, specific industries, and the behavior of individual consumers and firms.

(i) Level of Analysis:

Macroeconomics analyzes national and global economic trends and their impact.

Microeconomics analyzes the behavior of individual agents within specific markets.

(ii) Variables:

Macroeconomics focuses on aggregated variables like GDP, inflation, and unemployment.

Microeconomics examines variables such as supply and demand, individual consumer preferences, and production costs.

(iii) Policy Objectives:

Macroeconomics aims to achieve overall economic stability, sustainable growth, and full employment.

Microeconomics focuses on optimizing individual market outcomes and efficiency within specific industries.

(iv) Timeframe:

Macroeconomics often deals with long-term trends and systemic issues affecting the economy.

Microeconomics can analyze both short-term market dynamics and long-term industry trends.

(v) Policy Tools:

Macroeconomics employs tools like monetary policy and fiscal policy to manage national economies.

Microeconomics is less concerned with policy tools but may consider regulatory policies within specific markets.

(vi) Interdependence:

Macroeconomics recognizes that individual markets and agents are interconnected, and changes in one sector can affect others.

Microeconomics often assumes that the impact of individual decisions is limited to the specific market being analyzed.

In essence, macroeconomics focuses on the broader aspects of the economy, addressing issues like economic growth, inflation, and unemployment at a national or global level. Microeconomics, on the other hand, delves into the interactions of individual consumers, firms, and markets, examining how their decisions shape prices, quantities, and resource allocation within specific sectors. Both branches are essential for understanding the complexities of economic systems and making informed economic policy decisions.

MODEL PAPER-2
FIFTH SEMESTER [B.TECH]
ECONOMICS FOR ENGINEERS [HS-301]

Q.1. Explain the following in short:

Q.1. (a) Advantages of Demand Forecasting.

Ans. Many small business owners mistakenly believe that an annual budget that projects revenues is comparable to projecting demand. But a more thorough analysis of the market will help determine the timing of sales, which can help you better plan production, financing, labour and marketing.

1. Increase Supply Chain Efficiency

Improvement in supply chain efficiency is a key reason to use demand forecasting in operations management. If you can forecast not only the amount of sales you'll have but also when they are likely to occur, you can better schedule your production, warehousing and shipping. This helps you plan scheduled maintenance shutdowns away from busy sales periods and have adequate materials and labor on hand throughout the year. When you know about a coming spike in demand, you can contact your suppliers to make sure they have enough materials to keep your production lines running. You also can contact customers who don't have time-sensitive inventory needs and ask them to accept orders earlier or later than a specific time when you know you'll be busy. This also helps you better manage your warehouse and shipping needs.

2. Improve Labour Management

Another one of the benefits of demand forecasting is an improvement on the labour management side of your operation. Having too few workers to handle a spike in sales orders can lead to slow order fulfilment, pushing your customers to find other vendors to fill their orders. Even if this is only a temporary problem, once a customer buys from another vendor, you might lose that customer for good. Adding inadequately trained workers can increase quality control problems and product returns or a loss of customers. Having too many workers idle wastes money, and each extra worker on your staff increases your risk of a workers' compensation claim.

3. Insure Adequate Cash Flow

Knowing the peaks and valleys of demand helps you better manage your cash flow, ensuring you have enough money on hand to pay bills. Poor cash flow management can lead to an inability to pay vendors and suppliers on time, creating a risky situation where they may cut you off. An inability to manufacture your product leads to an inability to supply your customers. Demand forecasting lets you reserve cash or negotiate bridge loans or credit terms in advance.

4. Create Accurate Budgeting

In addition to your master budget, you should create cash flow, overhead, manufacturing, labor and marketing budgets. The more accurately you can forecast demand, including the timing of your sales, the more accurate you can be with budgeting. If you have a flexible budget, such as tying marketing spending to sales, you can shift paid marketing efforts such as advertising and free marketing efforts such as a social media campaign between slow and busy periods.

Q.1. (b) National Income Accounting

Ans. National income accounting refers to the government bookkeeping system that measures the health of an economy, projected growth, economic activity, and

development during a certain period of time. It helps in assessing the performance of an economy and the flow of money in an economy. The double entry system principle of accounting is used to prepare the national income accounts.

National Income Accounting Equation

The national income equation represents the relationship between national income and the economy's expense, along with other attributes, as shown in the following equation:

$$Y = C + I + G + (X - M)$$

Where:

- Y – National income
- C – Personal consumption expenditure
- I – Private investment
- G – Government spending
- X – Exports
- M – Imports

National Income Accounting and Gross Domestic Product

Gross Domestic Product (GDP), Net National Product (NNP), Gross National Product (GNP), personal income, and disposable income are the important metrics determined by national income accounting.

However, the most commonly used measure of the economy is GDP. It is the cumulative value of products and services generated in an economy over a given period of time. Only the goods produced in the home country are included in the GDP, regardless of the nationality status of the company owners.

The gross domestic product figure may not represent the correct value, as some goods may not even make it to the market, which makes it difficult to determine the true value of the market. Nevertheless, GDP reasonably represents the national output. Other economic measures can be derived from GDP.

Q.1. (c) Price Discrimination

Ans. Price discrimination is a selling strategy that charges customers different prices for the same product or service based on what the seller thinks they can get the customer to agree to. In pure price discrimination, the seller charges each customer the maximum price they will pay. In more common forms of price discrimination, the seller places customers in groups based on certain attributes and charges each group a different price.

Price discrimination occurs when different consumers are charged different prices for the same product or service. Specifically, those who are willing to pay more will be charged a higher price whereas price-sensitive individuals will be charged less.

Price Discrimination Types

Price discrimination is a selling strategy that charges customers different prices for the same product or service based on what the seller thinks they can get the customer to agree to. In pure price discrimination, the seller charges each customer the maximum price they will pay.

(a) First-degree price discrimination

First-degree price discrimination is also known as perfect price discrimination. In this type of discrimination, producers charge their customers the maximum amount they are willing to pay and capture the entire consumer surplus.

(b) Second-degree price discrimination

Second-degree discrimination happens when the company charges prices based

the amounts or quantities consumed. A buyer making bulk purchases will receive a lower price compared to those purchasing a small quantity.

(c) **Third-degree price discrimination**

Third-degree price discrimination occurs when the company charges different prices for customers from different backgrounds or demographics.

Necessary conditions for price discrimination

Here are some conditions for price discrimination to occur:

1. **A degree of monopoly power:** The company must have sufficient market power in order to price discriminate. In other words, it needs to be a price maker.
2. **The ability to define customer segments:** The company must be able to separate the market based on customers' needs, characteristics, time, and location.
3. **The elasticity of demand:** Consumers must vary in the elasticity of their demand. For example, demand for air travel from low-income consumers is more price elastic. In other words, they will be less willing to travel when the price increases compared to wealthier people.
4. **Prevention of re-sale:** The company must be able to prevent its products from being resold by another group of customers.

Advantages and disadvantages of price discrimination

A firm only considers price discrimination when the profit of separating the market is greater than keeping it whole.

Advantages

- (i) **Brings more revenues for the seller:** Price discrimination gives the firm a chance to increase its profit more than when charging the same price for everyone. For many businesses, it's also a way to make up for losses during the peak seasons.
- (ii) **Lowers the price for some customers:** Some groups of customers such as older people or students can benefit from lower prices as a result of price discrimination.
- (iii) **Regulates the demand:** A company can utilise low pricing to encourage more purchases during the off-season and avoid crowdedness during the peak seasons.

Disadvantages

- (i) **Reduces consumer surplus:** price discrimination transfers the surplus from consumer to producer, thus reducing the benefit consumers can receive.
- (ii) **Lower product choices:** some monopolies can take advantage of price discrimination to capture a greater market share and establish a high barrier to entry. This limits the product choices on the market and results in lower economic welfare. In addition, lower-income consumers may not be able to afford the high prices charged by the companies.
- (iii) **Creates unfairness in society:** customers who pay a higher price are not necessarily poorer than those paying a lower price. For example, some working-class adults have less income than retired people.
- (iv) **Administration costs:** there are costs for businesses that carry out price discrimination. For example, the costs to prevent customers from reselling the product to other consumers.

Price discrimination exists to help businesses capture more consumer surplus and maximise their profits. The types of price discrimination vary wildly from charging the customers by their maximum willingness to pay, the quantities purchased, or their age and gender.

For many groups of customers, price discrimination provides a huge benefit as they can pay a lower price for the same product or service. However, there may be potential unfairness in society and high administration costs for firms to prevent re-selling among customers.

Q.1. (d) Long Run Average Cost Curve

Ans. The long run is the period of time when all costs are variable. The long run depends on the specifics of the firm in question—it is not a precise period of time. If you have a one-year lease on your factory, then the long run is any period longer than a year, since after a year you are no longer bound by the lease. No costs are fixed in the long run. A firm can build new factories and purchase new machinery, or it can close existing facilities. In planning for the long run, the firm will compare alternative production technologies.

In this context, technology refers to all alternative methods of combining inputs to produce outputs. It does not refer to a specific new invention like the tablet computer. The firm will search for the production technology that allows it to produce the desired level of output at the lowest cost. After all, lower costs lead to higher profits—at least if total revenues remain unchanged. Moreover, each firm must fear that if it does not seek out the lowest-cost methods of production, then it may lose sales to competitor firms that find a way to produce and sell for less.

Shapes of Long-Run Average Cost Curves

While in the short run firms are limited to operating on a single average cost curve, in the long run when all costs are variable, they can choose to operate on any average cost curve. Thus, the long-run average cost (LRAC) curve is actually based on a group of short-run average cost (SRAC) curves, each of which represents one specific level of fixed costs. More precisely, the long-run average cost curve will be the least expensive average cost curve for any level of output. Figure shows how the long-run average cost curve is built from a group of short-run average cost curves. Five short-run-average cost curves appear on the diagram. Each SRAC curve represents a different level of fixed costs. For example, you can imagine $SRAC_1$ as a small factory, $SRAC_2$ as a medium factory, $SRAC_3$ as a large factory, and $SRAC_4$ and $SRAC_5$ as very large and ultra-large. Although this diagram shows only five SRAC curves, presumably there are an infinite number of other SRAC curves between the ones that are shown. This family of short-run average cost curves can be thought of as representing different choices for a firm that is planning its level of investment in fixed cost physical capital—knowing that different choices about capital investment in the present will cause it to end up with different short-run average cost curves in the future.

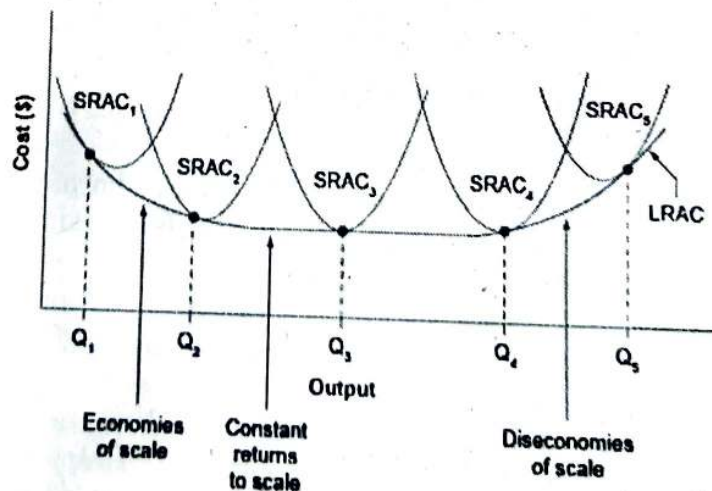


Fig. 1.

The long-run average cost curve shows the cost of producing each quantity in the long run, when the firm can choose its level of fixed costs and thus choose which short-run average costs it desires. If the firm plans to produce in the long run at an output of Q_3 , it should make the set of investments that will lead it to locate on $SRAC_3$, which

is choosing the level of fixed costs at $SRAC_0$ or $SRAC_1$. At $SRAC_0$, the level of fixed costs is too low for producing Q_0 at lowest possible cost, and producing Q_0 would require adding a very high level of variable costs and make the average cost very high. At $SRAC_1$, the level of fixed costs is too high for producing Q_0 at lowest possible cost, and again average cost would be very high as a result.

Q.1. (c) Law of Diminishing Marginal Utility.

Ans. The Law of Diminishing Marginal Utility is a fundamental concept in economics that explains how the satisfaction or utility derived from consuming additional units of a good or service decreases as consumption increases, while keeping other factors constant. In simpler terms, it states that the more of a particular product a person consumes, the less additional satisfaction they will derive from consuming each additional unit.

Marginal Utility: Marginal utility refers to the additional satisfaction or benefit gained from consuming one more unit of a good or service. It quantifies the change in utility resulting from a change in consumption.

Diminishing Returns: The law suggests that as a person consumes more of a good, the marginal utility derived from each additional unit tends to decrease. This happens because people tend to allocate the best units of a good to their most desired uses first. As they consume more, they might have to use the remaining units for less preferred purposes.

Example: Consider a person eating slices of pizza. The first slice might provide a lot of satisfaction, as it fulfills hunger. The second slice adds to the enjoyment but not as much as the first. As more slices are consumed, the incremental satisfaction from each additional slice becomes smaller, eventually reaching a point where eating more slices might lead to discomfort or even decreased utility.

Law of Equi-Marginal Utility: This related concept suggests that rational consumers allocate their spending or consumption in a way that the marginal utility per dollar spent is equal across different goods. In other words, consumers seek to maximize their overall satisfaction by considering the relative utility gained from each additional unit of various goods.

Real-World Applications: The Law of Diminishing Marginal Utility has implications for various economic behaviors, including consumption patterns, pricing strategies, and resource allocation. It helps explain why people diversify their consumption and why price discounts on larger quantities are often used to encourage consumption.

Exceptions and Considerations: While the law generally holds true for most goods and services, there are exceptions. Some goods, like addictive substances, might not follow this law due to psychological factors. Additionally, the law assumes that other factors remain constant, such as income, preferences, and quality of the good.

Understanding the Law of Diminishing Marginal Utility is crucial for economists, businesses, and policymakers as it helps explain consumer behavior, demand curves, and the factors influencing consumption choices. It also provides insights into the trade-offs people make when deciding how to allocate their limited resources among various goods and services.

Q.1. (d) Fiscal Policy.

Ans. Fiscal policy is the use of government spending and taxation to influence the economy. When the government decides on the goods and services it purchases, the transfer payments it distributes, or the taxes it collects, it is engaging in fiscal

policy. The primary economic impact of any change in the government budget is felt by particular groups—a tax cut for families with children, for example, raises their disposable income. Discussions of fiscal policy, however, generally focus on the effect of changes in the government budget on the overall economy. Although changes in taxes or spending that are “revenue neutral” may be construed as fiscal policy—and may affect the aggregate level of output by changing the incentives that firms or individuals face—the term “fiscal policy” is usually used to describe the effect on the aggregate economy of the overall levels of spending and taxation, and more particularly, the gap between them.

Fiscal policy factors and tools

Economic factors

The success of the economy is commonly measured by a few factors, including GDP. Another factor is aggregate demand, which is the sum of goods and services produced by a nation purchased at a certain price point. The aggregate demand curve dictates that at lower price levels, more goods and services are demanded, while there is less demand at higher price points.

Fiscal policy affects these measurements, with the goal of increasing GDP and aggregate demand in a sustainable manner. This happens by changing three factors.

1. Business tax policy: Taxes that businesses pay to the government affects its profits and investment spending. Lowering taxes increases both aggregate demand and business investment opportunities.

2. Government spending: Aggregate demand is increased by the government's own spending.

3. Individual taxes: Taxes on individuals – such as income tax – affect their personal income and how much they can spend, injecting more money back into the economy.

Fiscal policy typically needs to be altered when an economy is running low on aggregate demand and unemployment levels are high.

Policy tools

The two main tools of fiscal policy are taxes and spending. Taxes influence the economy by determining how much money the government has to spend in certain areas and how much money individuals should spend. For example, if the government is trying to spur consumer spending, it can decrease taxes. A cut in taxes provides families with extra cash, which the government hopes will in turn be spent on goods and services, thus spurring the economy as a whole.

Spending is used as a tool for fiscal policy to drive government money to specific sectors needing an economic boost. Whoever receives those dollars will have extra money to spend – and, as with taxes, the government hopes that money will be spent on other goods and services.

Q.1. (g) Monopolistic completion.

Ans. Monopolistic competition lies in-between. It involves many firms competing against each other, but selling products that are distinctive in some way. Examples include stores that sell different styles of clothing, restaurants or grocery stores that sell different kinds of food and even products like golf balls or beer that may be at least somewhat similar but differ in public perception because of advertising and brand names. Firms producing such products must also compete with other styles, flavours and brand names. The term “monopolistic competition” captures this mixture of mini-monopoly and tough competition.

Differentiated Products

A firm can try to make its products different from those of its competitors in several ways: physical aspects of the product, selling location, intangible aspects of the product, and perceptions of the product. Products that are distinctive in one of these four ways are called differentiated products.

Physical aspects of a product include all the phrases you hear in advertisements: such as an unbreakable bottle, nonstick surface, freezer-to-microwave, non-shrink, extra spicy, newly redesigned for your comfort. The location of a firm can also create a difference between producers. For example, a gas station located at a busy intersection can probably sell more gas than one located on a small side-road. A supplier to an automobile manufacturer may find that it is advantageous to locate near the car factory.

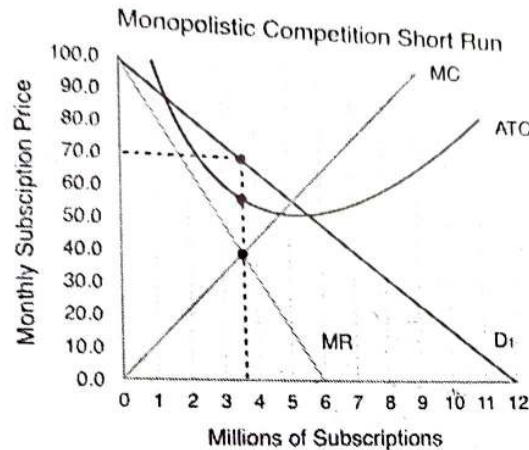


Fig. 2.

The monopolistically competitive firm decides on its profit-maximizing quantity and price similar to the way that a monopolist does. Since they face a downward sloping demand curve, the same considerations about how elasticity affects revenue are relevant, and the firm will maximize profits where $MR = MC$ when $P > MR$.

Q.2. What do you understand by market equilibrium price. Explain

Ans. Introduction

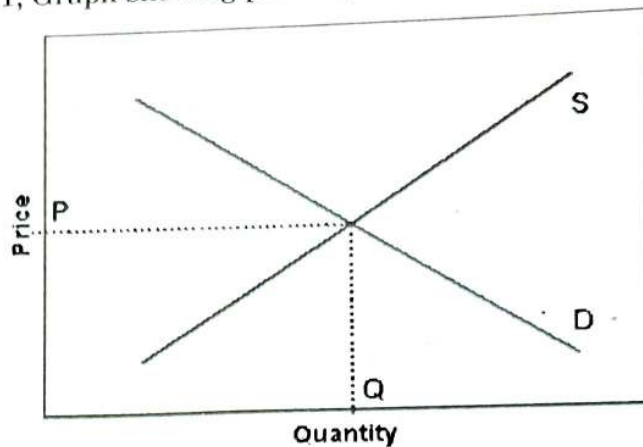
Price is dependent on the interaction between demand and supply components of a market. Demand and supply represent the willingness of consumers and producers to engage in buying and selling. An exchange of a product takes place when buyers and sellers can agree upon a price.

This section of the Agriculture Marketing Manual explains price in a competitive market. When imperfect competition exists, such as with a monopoly or single selling firm, price outcomes may not follow the same general rules.

Equilibrium price

When a product exchange occurs, the agreed upon price is called an equilibrium price, or a market clearing price. Graphically, this price occurs at the intersection of demand and supply as presented in Fig. 3

In Fig. 3, both buyers and sellers are willing to exchange the quantity Q at the price P . At this point, supply and demand are in balance. Price determination depends equally on demand and supply.

Image 1. Figure 1, Graph showing price equilibrium curves**Fig.3.** Price Equilibrium

It is truly a balance of the market components. To understand why the balance must occur, examine what happens when there is no balance, such as when market price is below that shown as P in Fig. 3.

At any price below P , the quantity demanded is greater than the quantity supplied. In such a situation, consumers would clamour for a product that producers would not be willing to supply; a shortage would exist. In this event, consumers would choose to pay a higher price in order to get the product they want, while producers would be encouraged by a higher price to bring more of the product onto the market.

The end result is a rise in price, to P , where supply and demand are in balance. Similarly, if a price above P were chosen arbitrarily, the market would be in surplus with too much supply relative to demand. If that were to happen, producers would be willing to take a lower price in order to sell, and consumers would be induced by lower prices to increase their purchases. Only when the price falls would balance be restored.

A market price is not necessarily a fair price, it is merely an outcome. It does not guarantee total satisfaction on the part of buyer and seller. Typically, some assumptions about the behaviour of buyers and sellers are made, which add a sense of reason to market price. For example, buyers are expected to be self-interested and, although they may not have perfect knowledge, at least they will try to look out for their own interests. Meanwhile, sellers are considered to be profit maximizers. This assumption limits their willingness to sell to within a price range, high to low, where they can stay in business.

Change in equilibrium price

When either demand or supply shifts, the equilibrium price will change. This section on understanding supply factors explains why a market component may move. The examples below show what happens to price when supply or demand shifts occur.

Example 1: Unusually good weather increases output

When a bumper crop develops, supply shifts outward and downward, shown as S_2 in Fig. 4, more product is available over the full range of prices. With no immediate change in consumers' willingness to buy crops, there is a movement along the demand curve to a new equilibrium. Consumers will buy more but only at a lower price. How much the price must fall to induce consumers to purchase the greater supply depends upon the elasticity of demand.

Image 2. Figure 4, Graph showing movement along demand curve

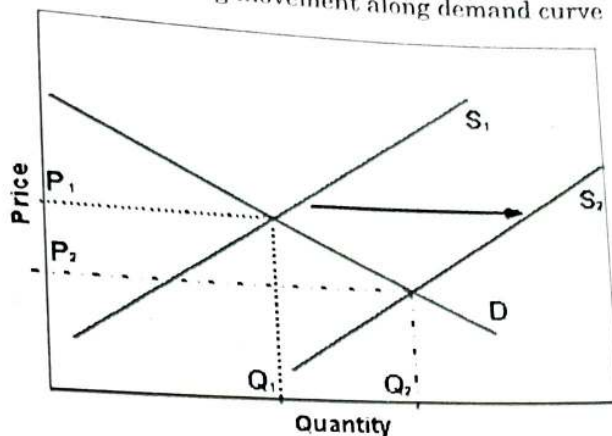


Fig. 4. Movement Along Demand Curve

In Fig. 4, price falls from P_1 to P_2 if a bumper crop is produced. If the demand curve in this example was more vertical (more inelastic), the price-quantity adjustments needed to bring about a new equilibrium between demand and the new supply would be different.

To understand how elasticity of demand affects the size of adjustment in prices and quantities when supply shifts, try drawing the demand curve (or line) with a slope more vertical than that depicted in Fig 4. Then compare the size of price-quantity changes in this with the first situation. With the same shift in supply, equilibrium change in price is larger when demand is inelastic than when demand is more elastic.

The opposite is true for quantity. A larger change in quantity will occur when demand is elastic compared with the quantity change required when demand is inelastic.

Example 2: Consumers lower their preference for beef

A decline in the preference for beef is one of the factors that could shift the demand curve inward or to the left, as seen in Fig.5.

Image 3. Figure 5. Graph showing movement along supply curve

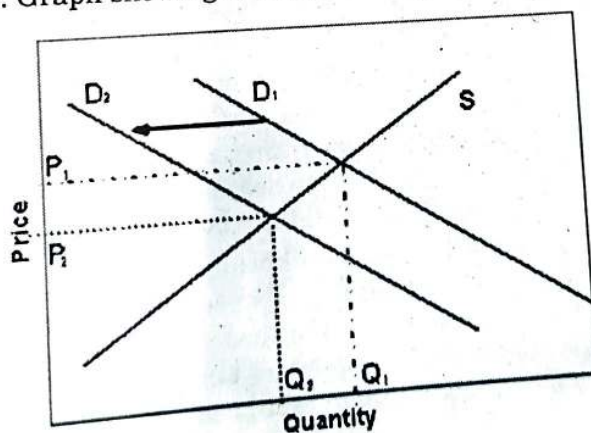


Fig. 5. Movement Along Supply Curve

With no immediate change in supply, the effect on price comes from a movement along the supply curve. An inward shift of demand causes price to fall and also the quantity exchanged to fall. The amount of change in price and quantity, from one equilibrium to another, is dependent upon the elasticity of supply.

Imagine that supply is almost fixed over the time period being considered. That is, draw a more vertical supply curve for this shift in demand. When demand shifts from D_1 to D_2 on a more vertical supply curve (inelastic supply) almost all the adjustment to a new equilibrium takes place in the change in price.

Price stability

Two forces contribute to the size of a price change: the amount of the shift and the elasticity of demand or supply. For example, a large shift of the supply curve can have a relatively small effect on price if the corresponding demand curve is elastic. That would show up in Example 1 above, if the demand curve is drawn flatter (more elastic).

In fact, the elasticity of demand and supply for many agricultural products are relatively small when compared with those of many industrial products. This inelasticity of demand has led to problems of price instability in agriculture when either supply or demand shifts in the short-term.

Price level

The two examples above focus on factors that shift supply or demand in the short-term. However, longer-term forces are also at work, which shift demand and supply over time. One particular supply shifter is technology. A major effect of technology in agriculture has been to shift the supply curve rapidly outward by reducing the costs of production per unit of output.

Technology has had a depressing effect on agricultural prices in the long-term since producers are able to produce more at a lower cost. At the same time, both population and income have been advancing, which both tend to shift demand to the right. The net effect is complex, but overall the rapidly shifting supply curve coupled with a slow moving demand has contributed to low prices in agriculture compared to prices for industrial products.

At various levels of a market, from farm gate to retail, unique supply and demand relationships are likely to exist. However, prices at different market levels will bear some relationship to each other. For example, if hog prices decline, it can be expected that retail pork prices will decline as well. This price adjustment is more likely to happen in the long-term once all participants have had time to adjust their behaviour.

In the short-term, price adjustments may not occur for a variety of reasons. For example, wholesalers may have long-term contracts that specify the old hog price, or retailers may have advertised or planned a feature to attract customers.

An equilibrium price, also known as a market-clearing price, is the consumer cost assigned to some product or service such that supply and demand are equal, or close to equal. The manufacturer or vendor can sell all the units they want to move and the customer can access all the units they want to buy.

Dynamic pricing seeks to find equilibrium prices in response to marketplace changes, adjusting prices on the fly in response to supply and demand fluctuations. Other models, such as value-based pricing, seek to capitalize on intangible qualities or employ various tactics to manipulate demand and achieve a higher profit margin than calculated what can be realized through cost-based pricing.

Equilibrium prices tend to remain stable but events that affect supply and demand inevitably affect prices as well. Cost increases for raw materials, for example, may mean that a manufacturer cannot produce the same product without increasing prices or cutting into the profit margin. The result may be that the manufacturer produces fewer units and charges a higher price. Fewer consumers may purchase at the higher price but, because fewer units are available, an equilibrium price may be reached.

Q.3. What do you mean by Circular flow of economic activities. Explain**Ans.** Introduction to the Circular Flow of Economic Activity.

The all-pervasive economic problem is that of scarcity which is solved by three institutions (or decision-making agents) of an economy. They are households (or individuals), firms and government. They are actively engaged in three economic activities of production, consumption and exchange of goods and services. These decision-makers act and react in such a manner that all economic activities move in a circular flow.

Nature and Role in decision-making.

Households:

Households are consumers. They may be single-individuals or group of consumers taking a joint decision regarding consumption. They may also be families. Their ultimate aim is to satisfy the wants of their members with their limited budgets.

Households are the owners of factors of production—land, labour, capital and entrepreneurial ability. They sell the services of these factors and receive income in return in the form of rent, wages, and interest and profit respectively.

Firms:

The term firm is used interchangeably with the term producer in economics. The decision to manufacture goods and services is taken by a firm. For this purpose, it employs factors of production and makes payments to their owners. Just as household's consumer goods and services to satisfy their wants, similarly firms produce goods and services to make a profit.

The term 'firm' includes joint stock companies like DCM, TISCO etc., public enterprises like IOC, STC, etc., partnership concerns, cooperative societies, and even small and big trading shops which do not manufacture the commodities they sell.

Government:

The government plays a key role in all types of economic systems—capitalist, socialist and mixed. In a capitalist economy, the government does not interfere. It simply establishes and protects property rights. It sets standards for weights and measures, and the monetary system.

In a socialist economy, the role of the government is very extensive. It owns and regulates the entire production and consumption processes of the economy, and fixes prices of goods and services. In a mixed economy, the government strengthens the market system.

It removes its defects by regulating the activities of the private sector and by providing incentives to it. The government also uses resources to produce goods and services itself which are sold to households and firms. These decision-making agents take economic decisions to produce goods and services and to exchange them in order to consume them for satisfying the wants of the whole economy.

Production, consumption and exchange are the three main activities of the economy. Consumption and production are flows which operate simultaneously and are interrelated and interdependent. Production leads to consumption and consumption necessitates production.

In other words, production is a means (beginning) and consumption is the end of all economic activities. Both production and consumption, in turn, depend upon exchange. Thus these two flows are interrelated and interdependent through exchange.

The Circular Flow in a Two-Sector Economy:

In a simplified economy with only two types of economic agents, households or consumers and business firms, the circular flow of economic activity is shown in Figure

10. Consumers and firms are linked through the product market where goods and services are sold. They are also linked through the factor market where the factors of production are sold and bought.

Consumers and firms have a dual role, and exchange with one another in two distinct ways:

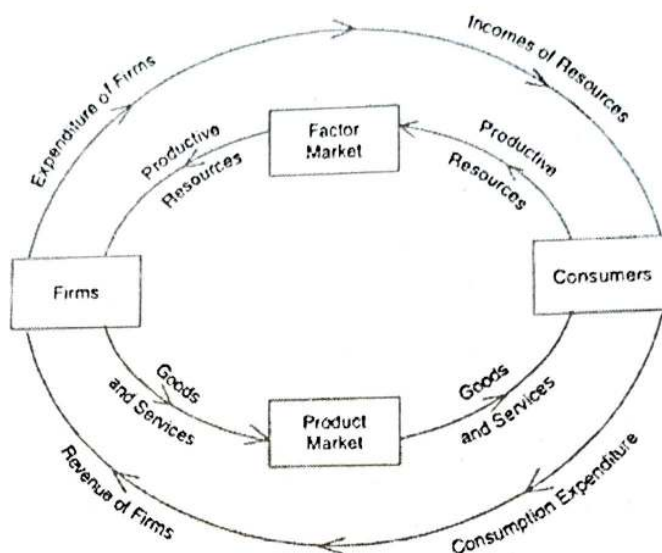


Fig. 6

(1) Consumers or households own all the factors of production, that is, land, labour, capital and entrepreneurship, which are also called productive resources. They sell them to firms for producing goods and services.

In the diagram, the sale of goods and services by firms to consumers in the product market is shown in the lower portion of the inner circle from left to right; and the sale of their services to firms by households or consumers in the factor market is shown in the upper portion of the inner circle from right to left. These are the real flows of goods and services from firms to consumers which are linked with productive resources from consumers to firms through the medium of exchange or barter.

(2) In a modern economy, exchange takes place through financial flows which move in the reverse direction to the "real" flows. The purchase of goods and services in the product market by consumers is their consumption expenditure which becomes the revenue of the firms and is shown in the outer circle of the lower portion from right to left in the diagram.

The expenditure of firms in buying productive resources in the factor market from the consumers becomes the incomes of households, which is shown in the outer circle of the upper portion from left to right in the diagram.

The Circular Flow in a Three-Sector Economy:

So far we have been working on the circular flow of a two-sector model of an economy. To this we add the government sector so as to make it a three-sector closed model of circular flow of economic activity. For this, we add taxes and government purchases (or expenditure) in our presentation.

Taxes are outflows from the circular flow and government purchases are inflows into the circular flow. The circular flow in a three-sector economy is illustrated in Figure 7.

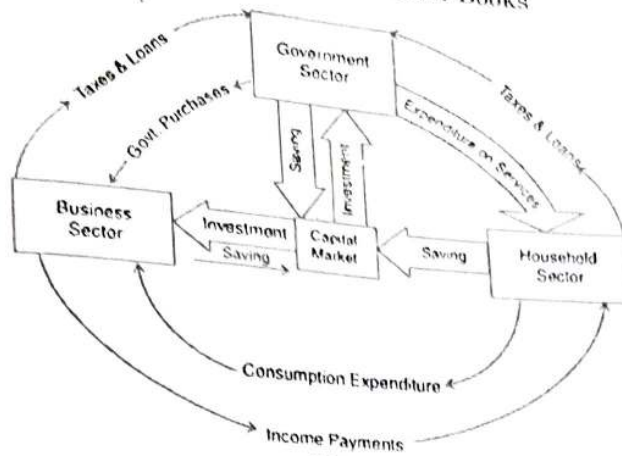


Fig. 7

First, take the circular flow between the household sector and the government sector. Taxes in the form of personal income tax and commodity taxes paid by the household sector are outflows (or leakages) from the circular flow. But the government purchases the services of the households, makes transfer payments in the form of old age pensions, unemployment relief, sickness benefit, etc., and also spends on them to provide certain social services like education, health, housing, water, parks and other facilities.

All such expenditures by the government are inflows (injections) into the circular flow. Next take the circular flow between the business sector and the government sector. All types of taxes paid by the business sector to the government are leakages from the circular flow.

On the other hand, the government purchases all its requirements of goods of all types from the business sector, gives subsidies and makes transfer payments to firms in order to encourage their production. These government expenditures are injections into the circular flow.

Now we take the household, business and government sectors together to show their inflows and outflows in the circular flow. As already noted, taxes are a leakage from the circular flow. They tend to reduce consumption and saving of the household sector. Reduced consumption, in turn, reduces the sales and incomes of the firms.

On the other hand, taxes on business firms tend to reduce their investment and production. The government offsets these leakages by making purchases from the business sector and buying services of the household sector equal to the amount of taxes. Thus inflows (injections) equal outflows (leakages) in the circular flow.

Figure 7 shows that taxes flow out of the household and business sectors and go to the government. The government purchases goods from firms and also factors of production from households. Thus government purchases of goods and services are an injection in the circular flow and taxes are leakages in the circular flow.

If government purchase exceeds net taxes then the government will incur a deficit equal to the difference between the two, i.e., government expenditure and taxes. The government finances its deficit by borrowing from the capital market which receives funds from the household sector in the form of saving.

On the other hand, if net taxes exceed government purchases the government will have a budget surplus. In this case, the government reduces the public debt and supplies funds to the capital market which are received by the business sector.

The Circular Flow in a Four-Sector Economy:

So far the circular flow has been shown in the case of a closed economy. But the actual economy is an open one where foreign trade plays an important role. Exports are an injection or inflows into the circular flow of money. On the other hand, imports are leakages from the circular flow.

They are expenditures incurred by the household sector to purchase goods from foreign countries. These exports and imports in the circular flow are shown in Figure 8.

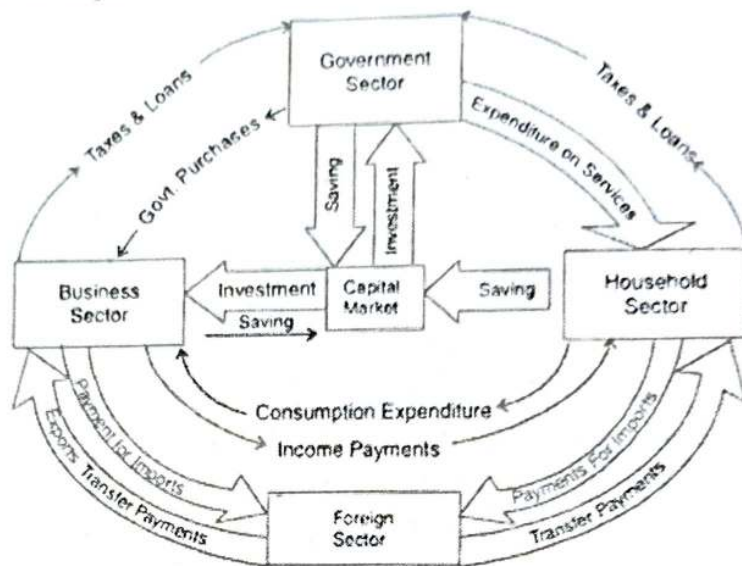


Fig. 8

Take the inflows and outflows of the household, business and government sectors in relation to the foreign sector. The household sector buys goods imported from abroad and makes payment for them which is a leakage from the circular flow of money. The householders' money receives transfer payments from the foreign sector for the services rendered by them in foreign countries.

On the other hand, the business sector exports goods to foreign countries and its receipts are an injection in the circular flow of money. Similarly, there are many services rendered by business firms to foreign countries such as shipping, insurance, banking, etc. for which they receive payments from abroad.

They also receive royalties, interests, dividends, profits, etc. for investments made in foreign countries. On the other hand, the business sector makes payments to the foreign sector for imports of capital goods, machinery, raw materials, consumer goods, and services from abroad. These are the leakages from the circular flow of money.

Like the business sector, modern governments also export and import goods and services, and lend to and borrow from foreign countries. For all exports of goods, the government receives payments from abroad.

Similarly, the government receives payments from foreigners when they visit the country as tourists and for receiving education, etc., and also when the government provides shipping, insurance and banking services to foreigners through the state-owned agencies.

Q.4. What is cost function and explain its types in detail.

Ans. The term 'cost of production' means expenses incurred in the production of a commodity. This refers to the total amount of money spent on the production of a commodity. The determinants of cost of production are: the size of plant, the level of production, the nature of technology used, the quantity of inputs used, managerial and labour efficiency. Thus the cost of production of a commodity is the

...the ... of ...

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 named in the above list of names.

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1. The first part of the document is a list of names and addresses of the members of the committee. The names are written in a cursive script, and the addresses are written in a more formal, printed style. The list includes names such as "John A. Smith", "Mary E. Jones", and "Robert L. Brown".

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3.2.2. *Phylogenetic analysis*

The first step in the process of making a good decision is to identify the problem. This involves understanding the situation, the people involved, and the resources available. Once the problem is identified, the next step is to generate possible solutions. This can be done by brainstorming or by consulting with others. After generating possible solutions, the next step is to evaluate each solution. This involves considering the pros and cons of each solution and determining which solution is the most feasible. Once a solution has been chosen, the final step is to implement the solution. This involves putting the solution into action and monitoring the results.

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1. The first part of the document is a letter from the author to the reader, explaining the purpose of the study and the methods used. The letter is dated 1998 and is addressed to the reader.

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1. The first step is to identify the problem or question that needs to be answered. This involves understanding the context and the specific requirements of the task.

the river causing water pollution, such a pollution results in tremendous health hazards which involve costs to the society as a whole

6. Fixed cost and variable cost

Fixed cost and variable cost are helpful in understanding the behaviour of costs over different levels of output.

Meaning of Fixed and Variable factors and costs

Fixed and variable factors are with reference to short run production function.

Short run is a period of time over which certain factors of production cannot be changed, and such factors are called fixed factors. The costs incurred on fixed factors are called fixed costs. The factors whose quantity can be changed in the short run are variable factors, and the costs incurred on variable factors are called variable costs.

Fixed costs are those which are independent of output, that is, they do not change with changes in output. These costs are a 'fixed' amount, which must be incurred by a firm in the short run whether the output is small or large. E.g. contractual rent, interest on capital invested, salaries to the permanent staff, insurance premia and certain taxes.

Variable costs are those costs, which are incurred on the employment of variable factors of production whose amount can be altered in the short run. Thus the total variable costs change with the level of output. It rises when output expands and falls when output contracts. When output is nil, variable cost becomes zero. These costs include payments such as wages of labour employed, prices of raw materials, fuel and power used and the transport costs.

7. Total cost

Total cost is the sum of total fixed cost and total variable cost. $TC = TFC + TVC$ where

TC = Total cost

TC = Total Fixed cost TVC = Total variable cost

It should be noted that total fixed cost is the same irrespective of the level of output. Therefore a change in total cost is influenced by the change in variable cost only.

The relationship between total fixed cost, total variable cost and total cost will be clear from the Figure.

Short run average cost curves

8. Average Fixed Cost (AFC)

The average fixed cost is the fixed cost per unit of output. It is obtained by dividing the total fixed cost by the number of units of the commodity produced.

Symbolically $AFC = TFC / Q$

Where AFC = Average fixed Cost TFC = Total Fixed cost

Q = number of units of output produced

Suppose for a firm the total fixed cost is Rs 2000 when output is 100 units, AFC will be $Rs\ 2000/100 = Rs\ 20$ and when output is 200 units, AFC will be $Rs\ 2000/200 = Rs\ 10/-$. Since total fixed cost is a constant quantity, average fixed cost will steadily fall as output increases; when output becomes very large, average fixed cost approaches zero.

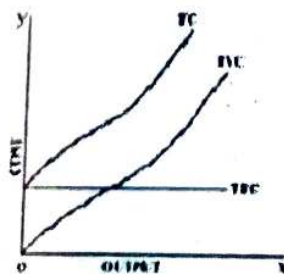


FIGURE 1

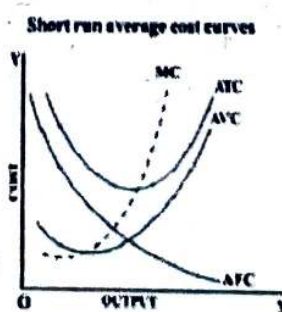


FIGURE 2

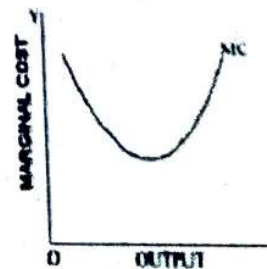


FIGURE 3

Average Variable cost (AVC): Average variable cost is the variable cost per unit of output. It is the total variable cost divided by the number of units of output produced.

$$AVC = TVC / Q$$

Where AVC = Average Variable Cost

TVC = Total Variable Cost

Q = number of units of output produced

Average variable cost curve is 'U' Shaped. As the output increases, the AVC will fall up to normal capacity output due to the operation of increasing returns. But beyond the normal capacity output, the AVC will rise due to the operation of diminishing returns.

Average Total Cost or Average Cost: Average total cost is simply called average cost which is the total cost divided by the number of units of output produced.

$$AC = TC / Q \text{ where } AC = \text{Average Cost } TC = \text{Total Cost}$$

Q = number of units of output produced

Average cost is the sum of average fixed cost and average variable cost. i.e.

$$AC = AFC + AVC$$

Q.5. (a) Explain the Exponential Technique.

Ans. Exponential smoothing was first suggested in the statistical literature without reference to previous work by Robert Goodell Brown in 1956 and then expanded by Charles C. Holt in 1957. Exponential smoothing is a broadly accurate principle for smoothing time series data using the exponential window function. The controlling input of the exponential smoothing calculation is defined as the smoothing factor or the smoothing constant.

As we know that, in the simple moving average, the past observations are weighted equally, exponential functions are used to assign exponentially decreasing weights over time. It is an easily learned and easily applied method for making some determination based on prior assumptions by the user, such as seasonality. Exponential smoothing is generally used for the analysis of time-series data.

Exponential Smoothing Formula

The simplest form of an exponential smoothing formula is given by:

$$s_t = \alpha x_t + (1 - \alpha)s_{t-1} = s_{t-1} + \alpha(x_t - s_{t-1})$$

Here,

s_t = smoothed statistic, it is the simple weighted average of current observation x_t

s_{t-1} = previous smoothed statistic

α = smoothing factor of data; $0 < \alpha < 1$

t = time period

If the value of the smoothing factor is larger, then the level of smoothing will reduce. Value of α close to 1 has less of a smoothing effect and give greater weight to recent changes in the data, while the value of α closer to zero has a greater smoothing effect and are less responsive to recent changes.

There is no official accurate procedure for choosing α . The statistician's judgment is used to choose an appropriate factor sometimes. Otherwise, a statistical technique may be used to optimize the value of α . For example, the method of least squares can be used to determine the value of α for which the sum of the quantities is minimized.

Exponential Smoothing Forecasting

Exponential smoothing is generally used to make short term forecasts, but longer-term forecasts using this technique can be quite unreliable. More recent observations given larger weights by exponential smoothing methods, and the weights decrease

exponentially as the observations become more distant. When the parameters describing the time series are changing slowly over time, then these methods are most effective.

Exponential Smoothing Methods

There are three main methods to estimate exponential smoothing. They are:

- Simple or single exponential smoothing
- Double exponential smoothing
- Triple exponential smoothing

(a) Simple or single exponential smoothing

If the data has no trend and no seasonal pattern, then this method of forecasting the time series is essentially used. This method uses weighted moving averages with exponentially decreasing weights.

The single exponential smoothing formula is given by:

$$s_t = \alpha x_t + (1 - \alpha)s_{t-1} = s_{t-1} + \alpha(x_t - s_{t-1})$$

(b) Double exponential smoothing

This method is also called as Holt's trend corrected or second-order exponential smoothing. This method is used for forecasting the time series when the data has a linear trend and no seasonal pattern. The primary idea behind double exponential smoothing is to introduce a term to take into account the possibility of a series showing some form of trend. This slope component is itself updated through exponential smoothing.

The double exponential smoothing formulas are given by:

$$S_1 = x_1$$

$$B_1 = x_1 - x_0$$

For $t > 1$,

$$s_t = \alpha x_t + (1 - \alpha)(s_{t-1} + b_{t-1})$$

$$\beta_t = \beta(s_t - s_{t-1}) + (1 - \beta)b_{t-1}$$

Here,

s_t = smoothed statistic, it is the simple weighted average of current observation x_t

s_{t-1} = previous smoothed statistic

α = smoothing factor of data; $0 < \alpha < 1$

t = time period

b_t = best estimate of trend at time t

β = trend smoothing factor; $0 < \beta < 1$

(c) Triple exponential smoothing

In this method, exponential smoothing applied three times. This method is used for forecasting the time series when the data has both linear trend and seasonal pattern. This method is also called Holt-Winters exponential smoothing.

Q.5. (b) Define Budget Constraint

Ans. A budget constraint is an economic term referring to the combined amount of items you can afford within the amount of income available to you.

The same principle of budget constraint can also be applied to time. If you are a manager, your employees only have a certain number of hours in a workday, which means you need to determine how much of their time they should spend on the various things they need to accomplish. The calculation may vary from week to week as your business priorities change and your employees' available time adjusts since some weeks include holidays or employees might take time off.

How do budget constraints work?

When calculating budget constraints, you normally have a number of things under consideration for which you are trying to budget. However, it's easier to understand how constraints work budget if you just consider two sets of items. You could spend your entire budget on item one, or you could spend it all on item two. Alternatively, you could buy a combination of some of item one and some of item two. The proportions of each item you purchase would be constrained by your budget.

If you are managing a department, calculating your budget constraint can help you determine whether the amount budgeted to you is adequate for your needs. Knowing how many things such as salaries, supplies and training materials you can afford within your budget constraint can help you determine if you need to request additional funding from your senior management.

Budget constraint equation

You can use the following equation to help calculate budget constraint:

$$(P1 \times Q1) + (P2 \times Q2) = m$$

In this equation, P1 is the cost of the first item, P2 is the cost of the second item and m is the amount of money available. Q1 and Q2 represent the quantity of each item you are purchasing. You could express this equation verbally by saying that the cost of the total number of X items added to the cost of the total number of Y items must equal the amount of money or income you have available.

If you draw this equation on a graph where the x-axis represents quantities of one item and the y-axis represents quantities of the other, it should plot a straight diagonal line sloping down from the left side to the right side. This line is called the budget line. Any point along the budget line indicates the quantities of each item you could purchase within your budget. If the price of one or both items changes, you would need to adjust this line accordingly.

Q.6. What is Elasticity of Demand? Explain the types of Elasticity of demand.

Ans. Elasticity of demand is an important variation on the concept of demand. Demand can be classified as elastic, inelastic or unitary.

An elastic demand is one in which the change in quantity demanded due to a change in price is large. An inelastic demand is one in which the change in quantity demanded due to a change in price is small.

The formula for computing elasticity of demand is:

$$\frac{(Q1 - Q2) / (Q1 + Q2)}{(P1 - P2) / (P1 + P2)}$$

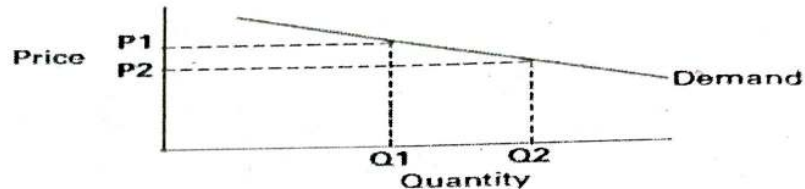
If the formula creates an absolute value greater than 1, the demand is elastic. In other words, quantity changes faster than price. If the value is less than 1, demand is inelastic. In other words, quantity changes slower than price. If the number is equal to 1, elasticity of demand is unitary. In other words, quantity changes at the same rate as price.

Elastic Demand

Elasticity of demand is illustrated in Figure 9. A change in price results in a large change in quantity demanded. An example of products with an elastic demand is consumer durables. These are items that are purchased infrequently, like a washing machine or an automobile, and can be postponed if price rises. For example, automobile rebates have been very successful in increasing automobile sales by reducing price.

Close substitutes for a product affect the elasticity of demand. If another product can easily be substituted for your product, consumers will quickly switch to the other product if the price of your product rises or the price of the other product declines. For example, beef, pork and poultry are all meat products. The declining price of poultry in recent years has caused the consumption of poultry to increase, at the expense of beef and pork. So products with close substitutes tend to have elastic demand.

Fig 9. Elastic Demand



An example of computing elasticity of demand using the formula is shown in Example 1. When the price decreases from \$10 per unit to \$8 per unit, the quantity sold increases from 30 units to 50 units. The elasticity coefficient is 2.25.

Example 1. Elastic Demand Example

$$P1 = \$10$$

$$P2 = \$8$$

$$Q1 = 30$$

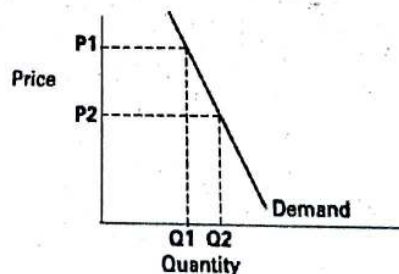
$$Q2 = 50$$

$$\frac{(Q1 - Q2) / (Q1 + Q2)}{(P1 - P2) / (P1 + P2)} = \frac{(50 - 30) / (50 + 30)}{(\$10 - \$8) / (\$10 + \$8)} = \frac{20 / 80}{\$2 / \$18} = \frac{1/4}{1/9} = \frac{1 \times 9}{4 \times 1} = \frac{9}{4} = 2.25$$

Inelastic Demand

Inelastic demand is shown in Figure 10. Note that a change in price results in only a small change in quantity demanded. In other words, the quantity demanded is not very responsive to changes in price. Examples of this are necessities like food and fuel. Consumers will not reduce their food purchases if food prices rise, although there may be shifts in the types of food they purchase. Also, consumers will not greatly change their driving behavior if gasoline prices rise.

Figure 10. Inelastic Demand



An example of computing inelasticity of demand using the formula above is shown in Example 2. When the price decreases from \$12 to \$6 (50%), the quantity of demand increases from 40 to only 50 (25%). The elasticity coefficient is .33.

Example 2. Inelastic Demand Example

$$P1 = \$12$$

$$P2 = \$6$$

$$Q1 = 40$$

$$Q2 = 50$$

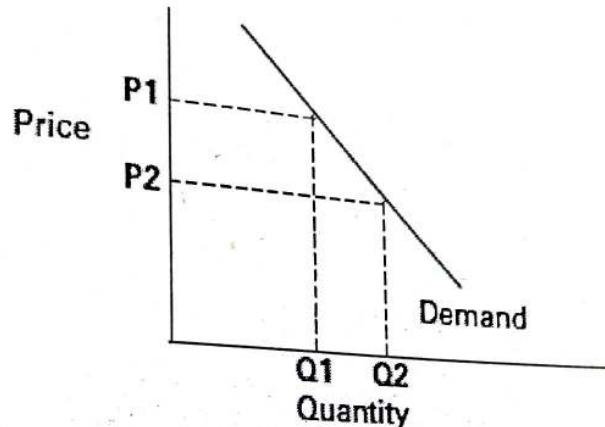
$$\frac{(Q_1 - Q_2) / (Q_1 + Q_2)}{(P_1 - P_2) / (P_1 + P_2)} = \frac{(50 - 40) / (50 + 40)}{(\$12 - \$6) / (\$12 + \$6)} = \frac{10/90}{\$6/\$18} = \frac{1/9}{1/3} = \frac{1 \times 3}{9 \times 1} = \frac{3}{9}$$

This does not mean that the demand for an individual producer is inelastic. For example, a rise in the price of gasoline at all stations may not reduce gasoline sales significantly. However, a rise of an individual station's price will significantly affect that station's sales.

Unitary Elasticity

If the elasticity coefficient is equal to one, demand is unitarily elastic as shown in Figure 11. For example, a 10% quantity change divided by a 10% price change is one. This means that a 1% change in quantity occurs for every 1% change in price.

Figure 11. Unitary Elasticity



Q.7. How Qualitative and Quantitative techniques can control the Inflation.

Ans. Quantitative Technique

Quantitative Technique is perhaps the simpler to define and identify.

The data produced are always numerical, and they are analysed using mathematical and statistical methods. If there are no numbers involved, then it's not quantitative research.

Some phenomena obviously lend themselves to quantitative analysis because they are already available as numbers. Examples include changes in achievement at various stages of education, or the increase in number of senior managers holding management degrees. However, even phenomena that are not obviously numerical in nature can be examined using quantitative methods.

The development of Likert scales and similar techniques mean that most phenomena can be studied using quantitative techniques.

This is particularly useful if you are in an environment where numbers are highly valued and numerical data is considered the 'gold standard'.

However, it is important to note that quantitative methods are not necessarily the most suitable methods for investigation. They are unlikely to be very helpful when you want to understand the detailed reasons for particular behaviour in depth. It is also possible that assigning numbers to fairly abstract constructs such as personal opinions risks making them spuriously precise.

Sources of Quantitative Data

The most common sources of quantitative data include:

Surveys, whether conducted online, by phone or in person. These rely on the same questions being asked in the same way to a large number of people;

Observations, which may either involve counting the number of times that a particular phenomenon occurs, such as how often a particular word is used in interviews, or coding observational data to translate it into numbers; and

Secondary data, such as company accounts.

Analysing Quantitative Data

There are a wide range of statistical techniques available to analyse quantitative data, from simple graphs to show the data through tests of correlations between two or more items, to statistical significance. Other techniques include cluster analysis, useful for identifying relationships between groups of subjects where there is no obvious hypothesis, and hypothesis testing, to identify whether there are genuine differences between groups.

Qualitative Technique

Qualitative research is any which does not involve numbers or numerical data.

It often involves words or language, but may also use pictures or photographs and observations.

Almost any phenomenon can be examined in a qualitative way, and it is often the preferred method of investigation in the UK and the rest of Europe; US studies tend to use quantitative methods, although this distinction is by no means absolute.

Qualitative analysis results in rich data that gives an in-depth picture and it is particularly useful for exploring how and why things have happened.

However, there are some pitfalls to qualitative research, such as:

If respondents do not see a value for them in the research, they may provide inaccurate or false information. They may also say what they think the researcher wishes to hear. Qualitative researchers therefore need to take the time to build relationships with their research subjects and always be aware of this potential.

Although ethics are an issue for any type of research, there may be particular difficulties with qualitative research because the researcher may be party to confidential information. It is important always to bear in mind that you must do no harm to your research subjects.

It is generally harder for qualitative researchers to remain apart from their work. By the nature of their study, they are involved with people. It is therefore helpful to develop habits of reflecting on your part in the work and how this may affect the research.

Sources of Qualitative Data

Although qualitative data is much more general than quantitative, there are still a number of common techniques for gathering it. These include:

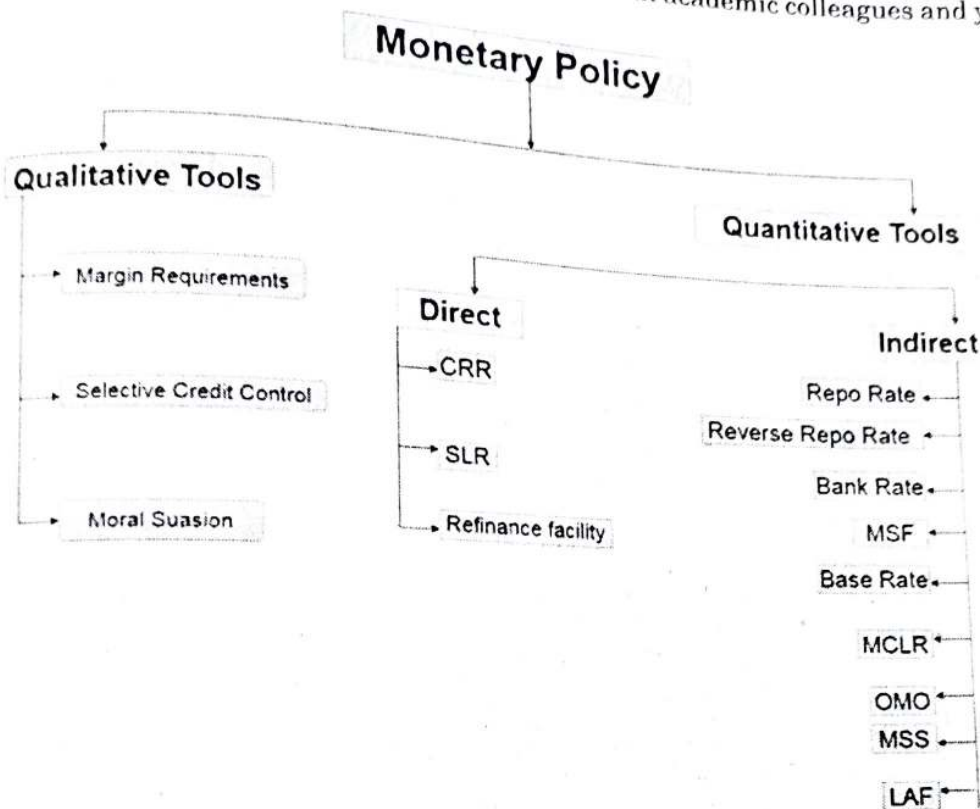
- Interviews, which may be structured, semi-structured or unstructured;
- Focus groups, which involve multiple participants discussing an issue;
- 'Postcards', or small-scale written questionnaires that ask, for example, three or four focused questions of participants but allow them space to write in their own words;
- Secondary data, including diaries, written accounts of past events, and company reports; and

Observations, which may be on site, or under 'laboratory conditions', for example, where participants are asked to role-play a situation to show what they might do.

Analysing Qualitative Data

Because qualitative data are drawn from a wide variety of sources, they can be radically different in scope.

There are, therefore, a wide variety of methods for analysing them, many of which involve structuring and coding the data into groups and themes. There are also a variety of computer packages to support qualitative data analysis. The best way to work out which ones are right for your research is to discuss it with academic colleagues and your supervisor.



Q.8. Write the characteristics of oligopoly markets and also define the level of equilibrium in this market. Explain

Ans. Oligopoly definition

Oligopoly occurs in industries where few but large leading firms dominate the market. Firms that are part of an oligopolistic market structure can't prevent other firms from gaining significant dominance in the market. However, as only a few firms have a significant share of the market, each firm's behaviour can have an impact on the other.

There must be a lower limit of two firms for a market structure to be considered oligopolistic, but there's no upper limit to how many firms are in the market. It is essential that there are a few and all of them combined have a significant share of the market, which is measured by the concentration ratio.

Oligopoly characteristics

The most important characteristics of oligopoly are interdependence, product differentiation, high barriers to entry, uncertainty, and price setters.

1. Firms are interdependent

As there are a few firms that have a relatively large portion of the market share, one firm's action impacts other firms. This means that firms are interdependent. There are two main methods through which a firm can influence the actions of other firms: by setting its price and output.

2. Product differentiation

When firms don't compete in terms of prices, they compete by differentiating their products. Examples of this include the automotive market, where one producer might add specific features that would help them acquire more customers. Although the car price might be the same, they are differentiated in terms of the features they have.

3. High barriers to entry

The market share acquired by the top companies in an industry becomes an obstacle for new companies to enter the market. The companies in the market use several strategies to keep other companies from entering the market. For instance, if firms collude, they choose the prices at a point where new companies can't sustain them. Other factors such as patents, expensive technology, and heavy advertising also challenge new entrants to compete.

4. Uncertainty

While companies in an oligopoly have perfect knowledge of their own business operations, they do not have complete information about other firms. Although firms are interdependent because they must consider other firms' strategies, they are independent when choosing their own strategy. This brings uncertainty to the market.

5. Price setters

Oligopolies engage in the practice of price-fixing. Instead of relying on the market price, firms set prices collectively and maximise their profits. Another strategy is to follow a recognised price leader; if the leader increases the price, the others will follow suit.

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MODEL PAPER-3
FIFTH SEMESTER [B.TECH]
ECONOMICS FOR ENGINEERS [HS-301]

Q.1. Explain the following in short:

Q.1. (a) Incremental Cost

Ans. An Incremental Cost is a cost resulting from additional expenses associated with the production of an additional unit or product. Incremental Cost is also called marginal Cost, it reflects changes that occur to the balance sheet of a company as a result of an addition to the unit of production. When a company produces one more unit of a product, the costs associated with this production are Incremental cost. An Incremental cost is important in managerial accounting; it is also useful when fixing the prices of goods and services.

Incremental cost is the extra cost associated with manufacturing one additional unit of production. It can be useful when formulating the price to charge a customer as part of a one-time deal to sell additional units. The concept can also be applied to cost reduction analysis, to enhance company profits. An incremental cost analysis only reviews those costs that will change as the result of a decision. All other costs are considered irrelevant to the decision.

When to Use Incremental Cost Analysis

It can be of interest to determine the incremental change in cost in a number of situations. For example, the incremental cost of an employee's termination includes the cost of additional benefits given to the person as a result of the termination, such as the cost of career counseling. Or, the incremental cost of shutting down a production line includes the costs to lay off employees, sell unnecessary equipment, and convert the facility to some other use. As a third example, the sale of a subsidiary includes the legal costs of the sale.

Q.1. (b) Regression Techniques.

Ans. Regression Analysis is a statistical process for estimating the relationships between the dependent variables or criterion variables and one or more independent variables or predictors. Regression analysis is generally used when we deal with a dataset that has the target variable in the form of continuous data. Regression analysis explains the changes in criteria in relation to changes in select predictors. The conditional expectation of the criteria is based on predictors where the average value of the dependent variables is given when the independent variables are changed. Three major uses for regression analysis are determining the strength of predictors, forecasting an effect, and trend forecasting.

What is the purpose to use Regression Analysis?

There are times when we would like to analyze the effect of different independent features on the target or what we say dependent features. This helps us make decisions that can affect the target variable in the desired direction. Regression analysis is heavily based on statistics and hence gives quite reliable results due to this reason only regression models are used to find the linear as well as non-linear relation between the independent and the dependent or target variables.

Types of Regression Techniques

Along with the development of the machine learning domain regression analysis techniques have gained popularity as well as developed manifold from just $y = mx + c$. There are several types of regression techniques, each suited for different types of data and different types of relationships. The main types of regression techniques are:

Linear regression

Linear regression is used for predictive analysis. Linear regression is a linear approach for modelling the relationship between the criterion or the scalar response and the multiple predictors or explanatory variables. Linear regression focuses on the conditional probability distribution of the response given the values of the predictors. For linear regression, there is a danger of overfitting. The formula for linear regression is:

Syntax:

$$y = 0x + b$$

where,

- 0 – It is the model weights or parameters
- b – It is known as the bias.

This is the most basic form of regression analysis and is used to model a linear relationship between a single dependent variable and one or more independent variables.

Polynomial Regression

This is an extension of linear regression and is used to model a non-linear relationship between the dependent variable and independent variables. Here as well syntax remains the same but now in the input variables we include some polynomial or higher degree terms of some already existing features as well. Linear regression was only able to fit a linear model to the data at hand but with polynomial features, we can easily fit some non-linear relationship between the target as well as input features.

Stepwise Regression

Stepwise regression is used for fitting regression models with predictive models. It is carried out automatically. With each step, the variable is added or subtracted from the set of explanatory variables. The approaches for stepwise regression are forward selection, backward elimination, and bidirectional elimination.

Decision Tree Regression

A Decision Tree is the most powerful and popular tool for classification and prediction. A Decision tree is a flowchart-like tree structure, where each internal node denotes a test on an attribute, each branch represents an outcome of the test, and each leaf node (terminal node) holds a class label. There is a non-parametric method used to model a decision tree to predict a continuous outcome.

Random Forest Regression

Random Forest is an ensemble technique capable of performing both regression and classification tasks with the use of multiple decision trees and a technique called Bootstrap and Aggregation, commonly known as bagging. The basic idea behind this is to combine multiple decision trees in determining the final output rather than relying on individual decision trees.

Random Forest has multiple decision trees as base learning models. We randomly perform row sampling and feature sampling from the dataset forming sample datasets for every model. This part is called Bootstrap.

Support Vector Regression (SVR)

Support vector regression (SVR) is a type of support vector machine (SVM) that is used for regression tasks. It tries to find a function that best predicts the continuous output value for a given input value.

SVR can use both linear and non-linear kernels. A linear kernel is a simple dot product between two input vectors, while a non-linear kernel is a more complex function that can capture more intricate patterns in the data. The choice of kernel depends on the data's characteristics and the task's complexity.

Ridge Regression

Ridge regression is a technique for analyzing multiple regression data. When multicollinearity occurs, least squares estimates are unbiased. This is a regularized linear regression model, it tries to reduce the model complexity by adding a penalty term to the cost function. A degree of bias is added to the regression estimates, and as a result, ridge regression reduces the standard errors.

Lasso Regression

Lasso regression is a regression analysis method that performs both variable selection and regularization. Lasso regression uses soft thresholding. Lasso regression selects only a subset of the provided covariates for use in the final model.

This is another regularized linear regression model, it works by adding a penalty term to the cost function, but it tends to zero out some features' coefficients, which makes it useful for feature selection.

Elastic Net Regression

Linear Regression suffers from overfitting and can't deal with collinear data. When there are many features in the dataset and even some of them are not relevant to the predictive model. This makes the model more complex with a too-inaccurate prediction on the test set (or overfitting). Such a model with high variance does not generalize on the new data. So, to deal with these issues, we include both L-2 and L-1 norm regularization to get the benefits of both Ridge and Lasso at the same time. The resultant model has better predictive power than Lasso. It performs feature selection and also makes the hypothesis simpler. The modified cost function for Elastic-Net Regression is given below:

where,

- $w(j)$ represents the weight for the j th feature.
- n is the number of features in the dataset.
- λ_1 is the regularization strength for the L1 norm.
- λ_2 is the regularization strength for the L2 norm.

Bayesian Linear Regression

As the name suggests this algorithm is purely based on Bayes Theorem. Because of this reason only we do not use the Least Square method to determine the coefficients of the regression model. So, the technique which is used here to find the model weights and parameters relies on features posterior distribution and this provides an extra stability factor to the regression model which is based on this technique.

Q.1. (c) Product Differentiation

Ans. Product differentiation is a process used by businesses to distinguish a product or service from other similar ones available in the market.

This tactic aims to help businesses develop a competitive advantage and define compelling, that set their product apart from competitors. In addition, organizations with multiple products in their portfolio may use differentiation to separate their various products from one another and prevent cannibalization.

This article will provide an overview of product differentiation and answer several common questions about this process.

Why is Product Differentiation Important?

In many industries, the barrier to entry has dropped significantly in recent years. As a side effect, these industries have seen substantial increases in competitive products. In increasingly crowded competitive landscapes, differentiation is a critical prerequisite for a product's survival.

What does your product or service deliver that the competition does not? Product differentiation helps your organization ensure that you deliver the unique value a product brings to its users. If an effort is put into a differentiation strategy, products risk blending in with a sea of competitors and not getting the market hold they need to keep going.

Who is Responsible for Product Differentiation?

While many schools of thought suggest differentiation is marketing's responsibility, that is not necessarily true. In fact, virtually every department within an organization can play a role in product differentiation.

Which teams are responsible for differentiation?

- Marketing
- Product Management
- Engineering
- Sales
- Support and Success

This is because any aspect of your product can be a differentiating factor. We usually think first of marketing because it's marketing that focuses on product positioning and is often the first touchpoint for a customer or prospect. However, differentiation is more than just how marketing positions the product—every single customer touchpoint is an opportunity for differentiation.

The answers to each of these questions could lead to a differentiating factor.

What are Types of Product Differentiation?

Several different factors can differentiate a product. However, there are three main categories of product differentiation. These include horizontal differentiation, vertical differentiation, and mixed differentiation.

Horizontal Differentiation

Horizontal differentiation refers to any differentiation that is not associated with the product's quality or price point. Instead, these products offer the same thing at the same price point. When making decisions regarding horizontally differentiated products, it often boils down to the customer's personal preference.

Examples of Horizontal Differentiation: Pepsi vs. Coca-Cola, bottled water brands, types of dish soap.

Vertical Differentiation

In contrast to horizontal differentiation, vertically differentiated products are extremely dependent on price. With vertically differentiated products, the price points and marks of quality are different. And, there is a general understanding that if all the options were the same price, there would be a clear winner for "the best."

Examples of Vertical Differentiation: Branded products vs. generics, A basic black shirt from Hanes vs. a basic black shirt from a top designer, the vehicle makes.

Mixed Differentiation

Also called "simple differentiation," mixed differentiation refers to differentiation based on a combination of factors. Often, this type of differentiation gets lumped in with horizontal differentiation.

Examples of Mixed Differentiation: Vehicles of the same class and similar price points from two different manufacturers.

Q.1. (d) Economies of Scale

Ans. Economies of scale are a reduction in costs to a business, which occurs when the company increases the production of their goods and becomes more efficient. This means that as businesses increase in size, it can lower their production costs and create a competitive advantage by either using those cost savings for increased profits or using the savings to lower the cost of their product to the consumer.

The importance of economies of scale:

Understanding economies of scale is important because of its effects on a business's production costs. Economies of scale create a competitive advantage for larger entities by putting out more production units and reducing their overall cost per unit.

As companies increase their production, they can spread out both their variable and fixed costs over a larger number of goods, lowering the per-unit cost of the product. When this happens, consumers may also benefit from reduced costs of goods.

Internal versus external economies of scale

There are two primary types of economies of scale:**1. Internal economies of scale**

Internal economies of scale result from a company being able to cut costs internally because of the size of the company or internal decisions made by managers and executive leadership. Internal economies of scale result from internal factors such as bulk purchasing, hiring more efficient and highly skilled managers and using technological advancements to lower production costs. These often create immediate effects for an organization, which can develop these effects for increased long-term growth.

2. External economies of scale

External economies of scale result from external factors outside the company's control, such as the industry, geographic area and the government. External economies of scale benefit cost reduction for the entire industry, not just for one company. These also typically create long-term effects for the entire industry, meaning everyone benefits in the long term and are harder to convert into short-term benefits.

Q.1. (e) Law of Equi-Marginal Utility.

Ans. Law of Equi-Marginal Utility explains the relation between the consumption of two or more products and what combination of consumption these products will give optimum satisfaction. Marginal Utility is the additional satisfaction gained by consuming one more unit of a commodity.

Marginal Utility

In simple words, it is the additional satisfaction gained by the consumption of one more unit of a commodity. It is the approximate change in the total utility resulting from a one-unit change in the consumption of the commodity. It decreases with each additional increase in the consumption of a good. You will think how?

Law of Diminishing Marginal Utility

This law explains the relation between utility and quantity of a commodity. It states that as consumption increases more and more, the marginal utility will be less and less.

Assumptions of Law of Equi-Marginal Utility

- Units of goods are homogenous.
- No time gap between the consumption of the different units.
- Tastes, fashion, preferences, and priorities remain unchanged.
- Consumer aims at maximum satisfaction.
- Consumer's income is fixed and limited.

Types of Marginal Utility

Based on the relationship between the total and the Marginal Utility, there are three types of marginal utility.

(a) Positive

The marginal utility is positive when the consumption of an additional unit of a product results in the increase in the total utility. Getting a coupon of free hair spa is its example.

(b) Negative

It is negative when the consumption of an additional unit of a product results in the decrease in the total utility. Taking more vitamin supplements or overtake of some medicine is its example.

(c) Zero

It is zero when the consumption of an additional unit of a product results in no change in the total utility. Getting two copies of the same novel is one of the examples of zero marginal utility.

Suppose we have data for the marginal and the total utility for different units of ice creams. Let us see the relation between the two.

Units	Total Utility	Marginal Utility
1	20	20
2	35	15
3	45	10
4	52	7
5	55	3
6	55	0
7	52	-2
8	47	-5
9	36	-11
10	20	-16

Importance of the Law

- This law is helpful in the field of production. A producer has limited resources and tries to get maximum profit.
- This law is helpful in the field of exchange. The exchange is of anything like some goods, wealth, trade, import, and export.
- It is applicable to public finance.
- The law is useful for workers in allocating the time between the work and rest.
- It is useful in case of saving and spending.
- It is useful to look for substitution in case of price rise.

Q.1. (f) Business Cycle

A business cycle is completed when it goes through a single boom and a single contraction in sequence. The time period to complete this sequence is called the length of the business cycle.

A boom is characterized by a period of rapid economic growth whereas a period of relatively stagnated economic growth is a recession. These are measured in terms of the growth of the real GDP, which is inflation-adjusted.

Stages of the Business Cycle

In the diagram above, the straight line in the middle is the steady growth line. The business cycle moves about the line. Below is a more detailed description of each stage in the business cycle:

1. Expansion

The first stage in the business cycle is expansion. In this stage, there is an increase in positive economic indicators such as employment, income, output, wages, profits, demand, and supply of goods and services. Debtors are generally paying their debts on time, the velocity of the money supply is high, and investment is high. This process continues as long as economic conditions are favorable for expansion.

2. Peak

The economy then reaches a saturation point, or peak, which is the second stage of the business cycle. The maximum limit of growth is attained. The economic indicators do not grow further and are at their highest. Prices are at their peak. This stage marks the reversal point in the trend of economic growth. Consumers tend to restructure their budgets at this point.

3. Recession

The recession is the stage that follows the peak phase. The demand for goods and services starts declining rapidly and steadily in this phase. Producers do not notice the decrease in demand instantly and go on producing, which creates a situation of excess supply in the market. Prices tend to fall. All positive economic indicators such as income, output, wages, etc., consequently start to fall.

4. Depression

There is a commensurate rise in unemployment. The growth in the economy continues to decline, and as this falls below the steady growth line, the stage is called depression.

5. Trough

In the depression stage, the economy's growth rate becomes negative. There is further decline until the prices of factors, as well as the demand and supply of goods and services, contract to reach their lowest point. The economy eventually reaches the trough. It is the negative saturation point for an economy. There is extensive depletion of national income and expenditure.

6. Recovery

After the trough, the economy moves to the stage of recovery. In this phase, there is a turnaround in the economy, and it begins to recover from the negative growth rate. Demand starts to pick up due to low prices and, consequently, supply begins to increase. The population develops a positive attitude towards investment and employment and production starts increasing.

Employment begins to rise and, due to accumulated cash balances with the bankers, lending also shows positive signals. In this phase, depreciated capital is replaced, leading to new investments in the production process. Recovery continues until the economy returns to steady growth levels.

This completes one full business cycle of boom and contraction. The extreme points are the peak and the trough.

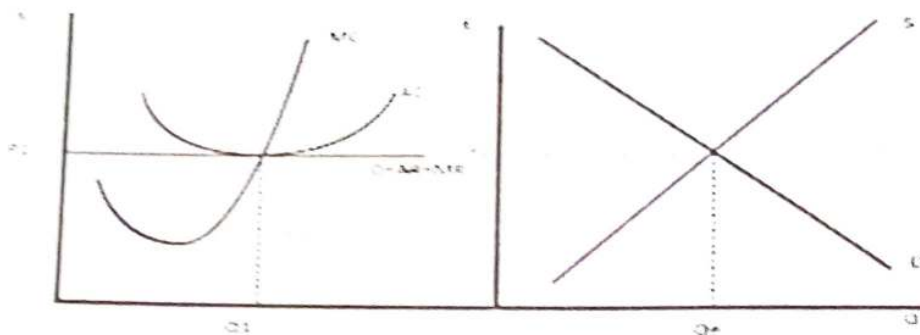
Q.1. (g) Perfect Competition.

Ans. Perfect competition is a type of market structure where all companies or firms are selling the same product, and because of having no control over their product prices, they tend to be price takers. In this market, consumers have full or perfect knowledge

about the product that is on sale. They know what firm charges what price for a specific product. There is a perfect mobility in terms of resources including labor, and there are no barriers to entry and exit involved for such firms.

How is the principle of Perfect Competition Used?

Perfect competition refers to a standard that enables the comparison of different market models. In theoretical language, it is totally the antonym of a monopoly where there is only one firm selling a product or service. Being the sole supplier and because of no competition, the firm becomes the price maker and has the ability to set any price for their products. They know that customers will buy their product no matter how high they set the prices. However, in perfect competition, there are lots of buyers and sellers for homogenous products, and this regulates the market demand and supply.



Firms manage to stay in a profitable position so as to keep their business going. Because of no barrier to entry, new firms can enter the market at any time, and affect the profitability of the existing firms. A market that is perfectly competitive has the following features:

- All companies sell homogenous products.
- All companies are price takers and not price makers.
- All companies enjoy lesser market share.
- Buyers already have knowledge about the type of product being sold followed by its prices.
- The firms can enter or exit the industry any time they want.

Q.2. What do you understand by market structure. Explain its types

Ans. The market structure consists of a number of firms that supply goods and services and the consumers who buy these goods and services. This helps to determine the level of production, consumption, and also competition. Depending on this, market structures are divided into concentrated markets and competitive markets.

Important features of market structure

The market structure consists of several features which we explain below.

1. Number of buyers and sellers

The main determinant of the market structure is the number of firms in the market. The number of buyers is also very important. Collectively, the number of buyers and sellers not only determines the structure and level of competition in a market but also influences the pricing and profit levels for the firms.

2. Barriers to entry and exit

Another feature that helps determine the type of market structure is the level of entry and exit. The easier it is for firms to enter and exit the market, the higher the level of competition. On the other hand, if the entry and exit are difficult, competition is much lower.

3. Perfect or imperfect information

The amount of information the buyers and sellers have in the markets also helps to determine the market structure. Information here includes product knowledge, production knowledge, prices, substitutes available, and the number of competitors for the sellers.

4. Nature of the product

What is the nature of a product? Are there any or close substitutes available for the product? Are the goods and services easily available in the market and are they identical and uniform? These are a few questions that we can ask to determine the nature of a product and therefore the market structure.

5. Price levels

Another key to identifying the type of market structure is to observe the price levels. A firm may be a price maker in one of the markets but a price taker in another. In some forms of markets, firms may have no control over the price, though in others there might be a price war.

Perfect competition

Perfect competition assumes that there are many suppliers and buyers for goods or services, and the prices are therefore competitive. In other words, firms are 'price-takers'.

These are the key features of perfect competition:

- There are a large number of buyers and sellers.
- Sellers/producers have perfect information.
- Buyers have perfect knowledge of the goods and services and the associated prices on the market.
- The firms have no barriers to entry and exit.
- The goods and services are homogeneous.
- No firm has supernormal profits due to low barriers to entry and exit.
- The firms are price takers.

However, this is a theoretical concept and such a market structure rarely exists in the real world. It is often used as a benchmark to assess the level of competition in other market structures.

Imperfect competition

Imperfect competition means there are many suppliers and/or many buyers in the market, which influences the demand and supply of the product thereby affecting the prices. Usually, in this form of market structure, the products sold are either heterogeneous or have some dissimilarities.

The imperfectly competitive market structures consist of the following types:

Monopolistic competition

Monopolistic competition refers to many firms supplying differentiated products. Firms may have a similar range of products, though not identical as in perfect competition. The differences will help them to set different prices from each other. The competition may be limited and firms compete to get buyers via lower prices, better discounts, or differentiated advertisements. The barrier to entry and exit is relatively low.

Q.3. What do you mean by Inflation. Explain its causes.

Ans. Inflation is a measure of how fast prices of goods and services are rising. If inflation is occurring, leading to higher prices for basic necessities such as food, it can have a negative impact on the overall economy.

Inflation can occur in nearly any product or service, including need-based expenses such as housing, food, medical care, and utilities, as well as want expenses, such as cosmetics, automobiles, and jewellery. Once inflation becomes prevalent throughout an economy, the expectation of further inflation becomes an overriding concern in the consciousness of consumers and businesses alike.

Inflation is the general increase in prices of goods and services within an economy over a period of time. It can be caused by various factors, both demand-side and supply-side. Here are some of the main causes of inflation:

1. Demand-Pull Inflation: This type of inflation occurs when the demand for goods and services exceeds their supply. Consumers have more money to spend, leading to increased demand for products. If supply struggles to keep up with this increased demand, prices tend to rise.

2. Cost-Push Inflation: This form of inflation is caused by increases in the cost of production, such as higher wages or raw material costs. When businesses face rising production costs, they often pass those costs onto consumers in the form of higher prices.

3. Built-In Inflation: Also known as wage-price inflation, this occurs when workers demand higher wages and businesses simultaneously raise prices to maintain their profit margins. This cycle can perpetuate inflation as workers' higher wages contribute to increased demand and costs for businesses.

4. Monetary Policy: Central banks, like the Federal Reserve in the United States, can influence inflation through their control of the money supply. If a central bank prints more money or lowers interest rates significantly, it can increase the amount of money circulating in the economy, potentially leading to demand-pull inflation.

5. Fiscal Policy: Government spending and taxation policies can also impact inflation. If a government increases its spending without matching revenue increases, it may lead to higher demand and potentially inflation. Similarly, tax cuts can put more money in consumers' hands, contributing to higher demand.

6. Supply Chain Disruptions: Natural disasters, geopolitical conflicts, and disruptions in global supply chains can reduce the supply of goods, leading to scarcity and subsequent price increases.

7. Imported Inflation: When a country heavily relies on imports, changes in exchange rates can affect the cost of imported goods. A weaker domestic currency can make imports more expensive, leading to higher prices for imported products.

8. Rapid Economic Growth: In developing economies, rapid economic growth can outpace the ability of producers to meet increased demand, resulting in demand-pull inflation.

9. Inflation Expectations: If people expect prices to rise in the future, they may adjust their behavior by purchasing goods and services now, contributing to increased demand and pushing prices up.

10. Energy and Commodity Prices: Fluctuations in the prices of energy (like oil) and key commodities can impact production costs across various industries, affecting overall price levels.

11. Global Factors: Events in the global economy, such as changes in international trade agreements, economic crises in other countries, and shifts in demand for goods and services worldwide, can influence domestic inflation.

In reality, inflation is often influenced by a combination of these factors, and their effects can vary depending on the economic context and the specific circumstances of each country. Central banks and policymakers strive to maintain stable inflation rates to support economic growth and stability.

Q.4. What is cost function and defines the short run cost function and long run cost function. Explains.

Ans. Cost in Short Run:

It may be noted at the outset that, in cost accounting, we adopt functional classification of cost. But in economics we adopt a different type of classification, viz., behavioural classification-cost behaviour is related to output changes.

In the short run the levels of usage of some input are fixed and costs associated with these fixed inputs must be incurred regardless of the level of output produced. Other costs do vary with the level of output produced by the firm during that time period.

The sum-total of all such costs-fixed and variable, explicit and implicit-is short-run total cost. It is also possible to speak of semi-fixed or semi-variable cost such as wages and compensation of foremen and electricity bill. For the sake of simplicity we assume that all short run costs to fall into one of two categories, fixed or variable.

Short-Run Total Cost:

A typical short-run total cost curve (STC) is shown in Fig. 1. This curve indicates the firm's total cost of production for each level of output when the usage of one or more of the firm's resources remains fixed.

When output is zero, cost is positive because fixed cost has to be incurred regardless of output. Examples of such costs are rent of land, depreciation charges, license fee, interest on loan, etc. They are called unavoidable contractual costs. Such costs remain contractually fixed and so cannot be avoided in the short run.

The only way to avoid such costs is by going into liquidation. The total fixed cost (TFC) curve is a horizontal straight line. Total variable is the difference between total cost and fixed cost. The total variable cost curve (TVC) starts from the origin, because such cost varies with the level of output and hence are avoidable. Examples are electricity tariff, wages and compensation of casual workers, cost of raw materials etc.

In Fig. 1 the total cost (OC) of producing Q units of output is total fixed cost OF plus total variable cost (FC).

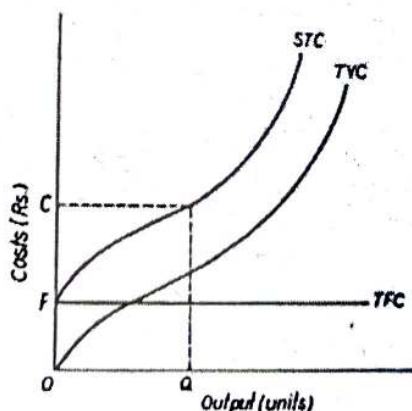


Fig. 1 Short-run Costs

early, variable cost and, therefore, total cost must increase with an increase in output. We also see that variable cost first increase at a decreasing rate (the slope of TVC decreases) then increase at an increasing rate (the slope of STC increases). This cost structure is accounted for by the law of Variable Proportions.

Average and Marginal Cost:

One can gain a better insight into the firm's cost structure by analysing the behaviour of short-run average and marginal costs. We may first consider average fixed cost (AFC).

ADVERTISEMENT:

Average fixed cost is total fixed cost divided by output,

$$\text{i.e., } AFC = TFC / Q$$

Since total fixed cost does not vary with output average fixed cost is a constant amount divided by output. Average fixed cost is relatively high at very low output levels. However, with gradual increase in output, AFC continues to fall as output increases, approaching zero as output becomes very large. In Fig. 2, we observe that the AFC curve takes the shape of a rectangular hyperbola.

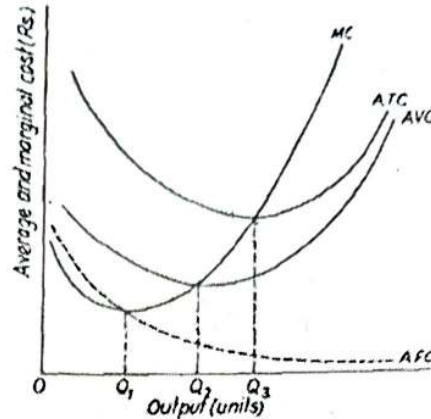


Fig. 2 Short-run average and marginal cost curves

We now consider average variable cost (AVC) which is arrived at by dividing total variable cost by output,

$$\text{i.e., } AVC = \frac{TVC}{Q}$$

In Fig. 2, AVC is a typical average variable cost curve. Average variable cost first falls, reaches a minimum point (at output level Q2) and subsequently increases.

The next important concept is one of average total cost (ATC).

It is calculated by dividing total cost by output,

$$\text{i.e., } ATC = \frac{TC}{Q}$$

$$\text{Alternatively, } TC = TFC + TVC$$

$$\begin{aligned} \text{and } ATC &= \frac{TFC}{Q} + \frac{TVC}{Q} \\ &= AFC + AVC \end{aligned}$$

It is, therefore, the sum of average fixed cost and average variable cost.

The ATC curve, illustrated, is U-shaped in Fig. 2 because the AVC cost curve is U-shaped. This is accounted for by the Law of Variable Proportions. It first declines, reaches a minimum (at Q3 units of output) and subsequently rises. The minimum point on ATC is reached at a larger output than at which AVC attains its minimum. This point can easily be proved.

$$ATC = AFC + AVC$$

We know that and that average fixed cost continuously falls over the whole range of output. Thus, ATC declines at first because both AFC and AVC are falling. Even when AVC begins to rise after Q2, the decrease in AFC continues to drive down ATC as output increases. However, an output of Q3 is finally reached, at which the increase in AVC overcomes the decrease in AFC, and ATC starts rising.

Since $ATC = AFC + AVC$, the vertical distance between average total cost and average variable cost measures average fixed cost. Since AFC declines over the entire range of output, AVC becomes closer and closer to ATC as output increases.

The marginal cost curve intersects AVC and ATC at their respective minimum points. This result follows from the definitions of the cost curves. If marginal cost curve lies below average variable cost curve the implication is clear: each additional unit of output adds less to total cost than the average variable cost.

Thus average variable cost has to fall. So long as MC is above AVC, each additional unit of output adds more to total cost than AVC. Thus, in this case, AVC must rise.

Thus when MC is less than AVC, average variable cost is falling. When MC is greater than AVC, average variable cost is rising. Thus MC must equal AVC at the minimum point of AVC. Exactly the same reasoning would apply to show MC crosses ATC at the minimum point of the latter curve.

(a) Short-run Cost Functions:

Summary of the Main Points All the important short-run cost relations may now be summed up:

The total cost function may be expressed as:

$TC = k + f(Q)$ where k is total fixed cost which is a constant, and $f(Q)$ is total variable cost which is a function of output.

$ATC = k/Q + f(Q)/Q = AFC + AVC$. Since k is a constant and Q gradually increases, the ratio k/Q falls. Hence the AFC curve is a rectangular hyperbola.

Here

$$MC = \frac{d(TC)}{dQ} = \frac{d}{dQ}(k) + \frac{d}{dQ}[f(Q)] = 0 + f'(Q)$$

where $f'(Q)$ is the change in TVC and may be called marginal variable cost (MVC). Thus, it is clear that MC refers to MVC and has no relation to fixed cost. Since business decisions are largely governed by marginal cost, and marginal costs have no relation to fixed cost, it logically follows costs do not affect business decisions.

(b) Relation between MC and AC:

There is a close relation between MC and AC. When AC is falling, MC is less than AC. This can be proved as follows:

When AC is falling,

$$\frac{d}{dQ}\left(\frac{TC}{Q}\right) < 0, \quad \text{or,} \quad \frac{Q \times \frac{dTC}{dQ} - TC}{Q^2} < 0,$$

$$\text{or,} \quad \frac{dTC}{dQ} - \frac{TC}{Q} < 0$$

$$\text{or,} \quad \frac{dTC}{dQ} - \frac{TC}{Q} < 0$$

$$\text{or,} \quad \frac{dTC}{dQ} < \frac{TC}{Q}$$

$$\text{or,} \quad MC < AC.$$

(c) Cost Elasticity:

On the basis of the relation between MC and AC we can develop a new concept, viz., the concept of cost elasticity. It measures the responsiveness of total cost to a small change in the level of output.

It can be expressed as:

$$E_c = \frac{\% \text{ change in TC}}{\% \text{ change in } Q} = \frac{\Delta TC / TC}{\Delta Q / Q}$$

$$= \frac{\Delta TC}{\Delta Q} \div \frac{TC}{Q} = MC \div AC$$

So it is the ratio of MC to AC.

The properties of the average and marginal cost curves and their relationship to each other are as described in Fig. 2. From the diagram the following relationships can be discovered.

(1) AFC declines continuously, approaching both axes asymptotically (as shown by the decreasing distance between ATC and AVC) and is a rectangular hyperbola.

(2) AVC first declines, reaches a minimum at Q_2 and rises thereafter. When AVC is at its minimum, MC equals AVC.

(3) ATC first declines, reaches a minimum at Q_3 , and rises thereafter. When ATC is at its minimum, MC equals ATC.

(4) MC first declines, reaches a minimum at Q_1 , and rises thereafter. MC equals both AVC and ATC when these curves are at their minimum values.

The lowest point of the AVC curve is called the shut (close)-down point and that of the ATC curve the break-even point. These two concepts will be discussed in the context of market structure and pricing. Finally, we see that MC lies below both AVC and ATC over the range in which these curves decline; contrarily, MC lies above them when they are rising.

Table 1 numerically illustrates the characteristics of all the cost curves. Column (5) shows that average fixed cost decreases over the entire range of output. Columns (6) and (7) depict that both average variable and average total cost first decrease, then increase, with average variable cost attaining a minimum at a lower output than that at which average total cost reaches its minimum. Column (8) shows that marginal cost per 100 units is the incremental increase in total cost and variable cost.

(1) Output Rs.	(2) Total cost Rs.	(3) Fixed cost Rs.	(4) Variable cost Rs.	(5) Average fixed cost Rs.	(6) Average variable cost Rs.	(7) Average total cost Rs.	(8) Marginal cost (per unit) Rs.
100	6,000	4,000	2,000	40.00	20.00	60.00	20.00
200	7,000	4,000	3,000	20.00	15.00	35.00	10.00
300	7,500	4,000	3,500	13.33	11.67	25.00	5.00
400	9,000	4,000	5,000	10.00	12.50	22.50	15.00
500	11,000	4,000	7,000	8.00	14.00	22.00	20.00
600	14,000	4,000	10,000	6.67	16.67	23.33	30.00
700	18,000	4,000	14,000	5.71	20.00	25.71	40.00
800	24,000	4,000	20,000	5.00	25.00	30.00	60.00
900	34,000	4,000	30,000	4.44	33.33	37.77	100.00
1,000	50,000	4,000	46,000	4.00	46.00	50.00	160.00

If we compare columns (6) and (8) we see that marginal cost (per unit) is below average variable and average total cost when each is falling and is greater than each when AVC and ATC are rising.

Long-Run Costs: The Planning Horizon:

We may recall from our discussion of production theory that the long run does not refer to 'some date in the future. Instead, the long run simply refers to a period of time during which all inputs can be varied.

Therefore, a decision has to be made by the owner and/or manager of the firm about the scale of operation, that is, the size of the firm. In order to be able to make this decision the manager must have knowledge about the cost of producing each relevant level of output. We shall now discover how to determine these long-run costs.'

Derivation of Cost Schedules from a Production Function:

For the sake of analysis, we may assume that the firm's level of usage of the inputs does not affect the input (factor) prices. We also assume that the firm's manager has already evaluated the production function for each level of output in the feasible range and has derived an expansion path.

For the sake of analytical simplicity, we may assume that the firm uses only two variable factors, labour and capital, that cost Rs. 5 and Rs. 10 per unit, respectively.

The characteristics of a derived expansion path are shown in Columns 1, 2 and 3 of Table 2. In column (1) we see seven output levels and in Columns (2) and (3) we see the optimal combinations of labour and capital respectively for each level of output, at the existing factor prices.

These combinations enable us to locate seven points on the expansion path.

Column (4) shows the total cost of producing each level of output at the lowest possible cost. For example, for producing 300 units of output, the least cost combination of inputs is 20 units of labour and 10 of capital. At existing factor prices, the total cost is Rs. 200. Here, Column (4) is a least-cost schedule for various levels of production.

In Column (5), we show average cost which is obtained by dividing total cost figures of Column (4) by the corresponding output figures of Column (1). Thus, when output is 100, average cost is $\text{Rs. } 120/100 = \text{Rs. } 1.20$. All other figures of Column (5) are derived in a similar way.

From column (5) we derive an important characteristic of long-run average cost: average cost first declines, reaches a minimum, then rises, as in the short-run. In Column (6) we show long-run marginal cost figures.

Each such figure is arrived at by dividing change in total cost by change in output. For example, when output increases from Rs. 100 to Rs. 200, the total cost increases from Rs. 120 to Rs. 140. Therefore, marginal cost (per unit) is $\text{Rs. } 20/100 = \text{Rs. } 0.20$. Similarly, when output increases from 600 to 700 units, MC per unit is $720-560/100 = 160/100 = 1.60$

Column (6) depicts the behaviour of per unit MC: marginal cost first decreases then increases, as in the short run.

We may now show the relationship between the expansion path and long-run cost graphically. In Fig. 3 two inputs, K and L, are measured along the two axes. The fixed factor price ratio is represented by the slope of the isocost lines $I_1I'_1$, $I_2I'_2$ and so on. Finally, the known production function gives us the isoquant map, represented by Q_1 , Q_2 and so forth.

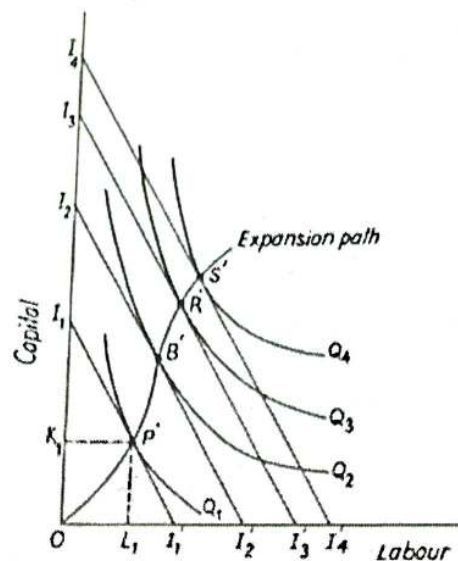


Fig. 3 The expansion path and long-run cost

From our earlier discussion of long-run production function we know that, when all inputs are variable (that is, in long-run), the manager will choose the least cost combinations of producing each level of output. In Fig. 3, we see that the locus of all such combinations is expansion path $OP'B'R'S'$.

Given the factor-price ratio and the production function (which is determined by the state of technology), the expansion path shows the combinations of inputs that enables the firm to produce each level of output at the lowest cost.

Table 2 : Derivation of long-run cost schedules

(1) Output (Units)	(2) Labour (Units)	(3) Least-usage of Capital of labour,	(4) Total cost at Rs. 5 per unit Rs. 10 per unit of capital	(5) Average cost	(6) Marginal cost (per unit)
100	11	7	Rs. 120	Rs. 120	Rs. 1.20
200	12	8	140	Rs. 0.70	Rs. 0.20
300	20	10	200	0.67	1.60
400	30	15	300	0.75	1.00
500	40	22	420	0.84	Rs. 1.20
600	52	30	560	0.93	1.40
700	60	42	720	Rs. 1.03	1.60

We may now relate this expansion path to a long-run total cost (LRTC) curve. Fig. 4 shows the 'least cost curve' associated with expansion path in Fig. 3. This least cost curve is the long-run total cost curve. Points P, B, R and S are associated with points P', B', R' and S' on the expansion path. For example, in Fig. 3 the least cost combination of inputs that can produce Q_1 is K_1 units of capital and L_1 units of labour.

Thus, in Fig. 4, minimum possible cost of producing Q_1 units of output is TC_1 , which is $K_1 + wL_1$, i.e., the price of capital (or the rate of interest) times K_1 , plus the price of labour (or the wage rate) times L_1 . Every other point on LRTC is derived in a similar way.

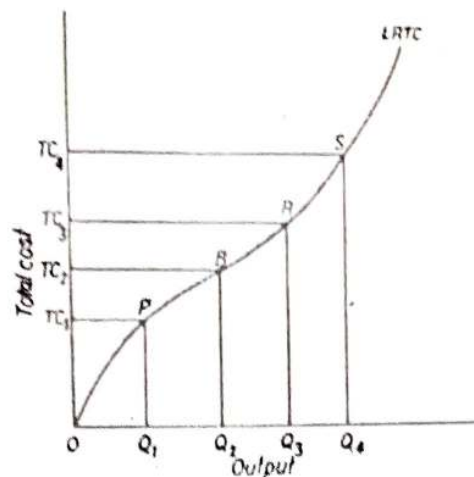


Fig. 4 Long-run total cost curve

Since the long run permits capital-labour substitution, the firm may choose different combinations of these two inputs to produce different levels of output. Thus, totally different production processes may be used to produce (say) Q_1 and Q_2 units of output at the lowest attainable cost.

On the basis of this diagram we may suggest a definition of the long run total cost. The time period during which even/thing (except factor prices and the state of technology or art of production) is variable is called the long run and the associated curve that shows the minimum cost of producing each level of output is called the long-run total cost curve.

The shape of the long-run total cost (LRTC) curve depends on two factors: the production function and the existing factor prices. Table 2 and Fig. 4 reflect two of the commonly assumed characteristics of long-run total costs. First, costs and output are directly related; that is, the LRTC curve has a positive slope. But, since there is no fixed cost in the long run, the long run total cost curve starts from the origin.

In the long run all factors of production are variable. No factor is fixed. Accordingly, the scale of production can be changed by changing the quantity of all factors of production.

Definition:

"The term returns to scale refers to the changes in output as all factors change by the same proportion." Koutsoyiannis

Returns to scale relates to the behaviour of total output as all inputs are varied and is a long run concept". Leibhafskey

Returns to scale are of the following three types:

1. Increasing Returns to scale.
2. Constant Returns to Scale
3. Diminishing Returns to Scale

Explanation:

In the long run, output can be increased by increasing all factors in the same proportion. Generally, laws of returns to scale refer to an increase in output due to increase in all factors in the same proportion. Such an increase is called returns to scale.

1. Increasing Returns to Scale: Increasing returns to scale or diminishing cost refers to a situation when all factors of production are increased, output increases at a higher rate. It means if all inputs are doubled, output will also increase at the faster rate than double. Hence, it is said to be increasing returns to scale. This increase is due to many reasons like division external economies of scale.

UNIT 1: THEORY OF FACTOR PROPORTIONING AND ALTERNATIVE FACTORS IN SCALE

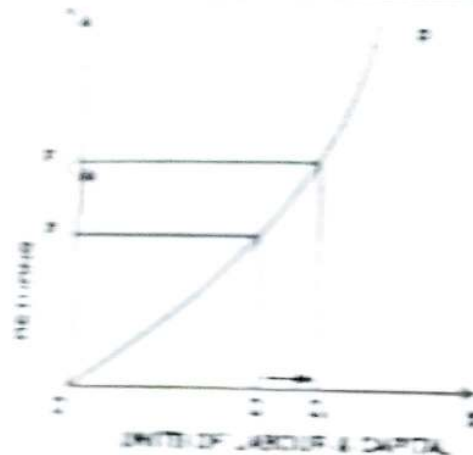


Fig. 1

Some of the factors are as follows:

(i) Technical and managerial indivisibility

Because the level of output is not a continuous variable, but is determined by the level of output, the level of output is not a continuous variable. These indivisible inputs are divided into two different levels of output. For example, an organization cannot use the level of the output of small scale of production.

Similarly, the organization cannot use half of a manager to achieve small scale of production. Due to the technical and managerial indivisibility, an organization needs a certain minimum quantity of machines and managers even if the level of production is small. Therefore, when there is an increase in output, there is a corresponding increase in the level of output.

(ii) Specialization

Because the high degree of specialization of labor and machinery helps in increasing the scale of production. The use of specialized labor and machinery helps in increasing the productivity of labor and capital per unit. This results in increasing returns to scale.

(iii) Increase in Dimensions

There is an increase in the level of output in the concept of dimensions according to the concept of dimensions. If the length and breadth of a room increases, then the area of the room also increases.

For example, length of a room increases from 15 to 30 and breadth increases from 10 to 20. The area of the room increases from 150 to 600. In such a case, the area of the room increases from 150 to 600, which is more than doubled.

1. Diminishing Returns to Scale

Diminishing returns to increasing costs refer to that production situation, where if the level of production is increased in a given proportion, output increases in a smaller proportion. I.e., if the level of production is doubled, output will be less than doubled. If a 10% increase in input is followed by 11% increase in output, then there is a diminishing return to scale.

The main cause of the diminishing return to scale is that physical and technical constraints are not met. These constraints are physical and technical constraints.

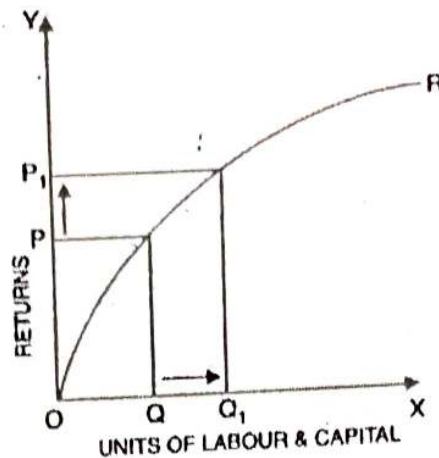


Fig. 6

3. Constant Returns to Scale:

Constant returns to scale or constant cost refers to the production situation in which output increases exactly in the same proportion in which factors of production are increased. In simple terms, if factors of production are doubled output will also be doubled.

In this case internal and external economies are exactly equal to internal and external diseconomies. This situation arises when after reaching a certain level of production, economies of scale are balanced by diseconomies of scale. This is known as homogeneous production function. Cobb-Douglas linear homogeneous production function is a good example of this kind. This is shown in Fig. 7. In figure 7, we see that increase in factors of production i.e. labour and capital are equal to the proportion of output increase. Therefore, the result is constant returns to scale.

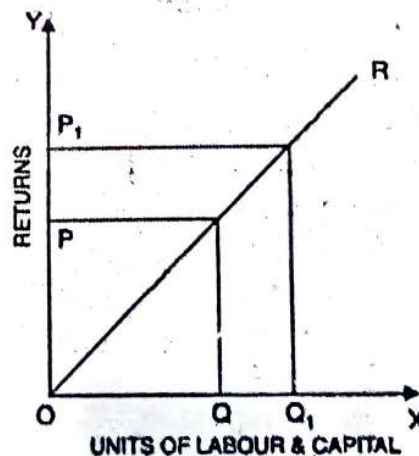


Fig. 7

Q.5. (a) Explain the Welfare Maximization

Ans. Welfare maximization, also known as social welfare maximization, refers to the goal of optimizing the overall well-being and satisfaction of individuals within a society. It is a central concept in economics and social policy, aimed at achieving the greatest possible benefit for the largest number of people. The idea behind welfare maximization is to create policies and make decisions that enhance the overall quality of life for everyone in a society.

In economic terms, welfare maximization involves maximizing the total utility or satisfaction that individuals derive from consuming goods and services. However, it's important to note that welfare maximization doesn't only focus on economic factors; it also considers social, environmental, and ethical dimensions of well-being.

Key points and considerations related to welfare maximization include:

(i) **Utility:** Utility is a measure of the satisfaction or well-being an individual derives from consuming goods and services. In welfare maximization, the goal is to allocate resources and make decisions in a way that maximizes the total utility across the population.

(ii) **Equity:** Welfare maximization often involves achieving a balance between efficiency (maximizing total utility) and equity (ensuring fair distribution of benefits). Policies that benefit a larger portion of the population and reduce inequality are generally favored.

Trade-offs: There are often trade-offs between different aspects of well-being. For example, policies that promote economic growth might have environmental costs. Welfare maximization requires assessing and optimizing these trade-offs.

(iii) **Externalities:** Externalities are unintended effects of economic activities on third parties who are not involved in those activities. Negative externalities (like pollution) can reduce overall welfare, while positive externalities (like education) can enhance it.

(iv) **Public Goods:** Public goods are non-excludable and non-rivalrous, meaning one person's consumption doesn't reduce their availability to others. Maximizing welfare involves providing and managing public goods efficiently.

(v) **Pareto Efficiency:** A situation is Pareto efficient if it's not possible to make any individual better off without making someone else worse off. Welfare maximization seeks to move society towards Pareto-efficient outcomes.

(vi) **Social Policies:** Welfare maximization guides the development and implementation of social policies, such as education, healthcare, income redistribution, and environmental regulations, with the aim of improving overall well-being.

(vii) **Interdisciplinary Approach:** Welfare maximization considers economic, sociological, psychological, and ethical factors. It often requires collaboration between economists, policymakers, social scientists, and ethicists.

(viii) **Dynamic Nature:** Societal needs and preferences change over time. Welfare maximization requires adaptability to evolving circumstances and challenges.

Welfare maximization is a complex and multifaceted concept, and its practical implementation can be challenging. Different societies have varying priorities, cultural norms, and values that influence how welfare is defined and pursued. Policymakers often use a combination of economic analyses, social indicators, and public input to make decisions that balance the diverse dimensions of welfare in a way that benefits the broader population.

Q.5. (b) Define Budget Line

Ans. The budget line is a graphical representation that shows the different combinations of two goods a consumer can afford to buy given their income and the prices of the goods. It is often used in conjunction with indifference curves to analyze consumer choices and preferences.

Here's how the budget line works within the context of indifference curves:

Budget Constraint: The budget line represents the constraint that a consumer faces when making purchasing decisions. It shows all the possible combinations of two goods (let's call them Good X and Good Y) that the consumer can afford to buy given a specific income and the prices of the two goods.

Income and Prices: The position of the budget line depends on the consumer's income and the prices of the goods. The equation of the budget line can be written as:

Income = (Price of Good X) $\times$ (Quantity of Good X) + (Price of Good Y) $\times$ (Quantity of Good Y)

Rearranging the equation, you can solve for the quantity of one good in terms of the other and the prices.

Graphical Representation: The budget line is usually depicted on a graph with Good X on the horizontal axis and Good Y on the vertical axis. The slope of the budget line is determined by the ratio of the prices of the two goods (Price of Good Y divided by Price of Good X). This slope represents the rate at which one good can be traded for another while keeping spending constant.

Consumer Choices: The budget line intersects with various indifference curves, which represent different levels of utility (satisfaction) for the consumer. The consumer's optimal choice is where the budget line is tangent to the highest attainable indifference curve. This point represents the consumer's most preferred combination of the two goods, given their budget constraint.

Changes in Income and Prices: Changes in income or the prices of the goods will cause the budget line to shift. An increase in income or a decrease in the price of a good will cause the budget line to pivot outward, allowing the consumer to afford more of both goods. Conversely, a decrease in income or an increase in the price of a good will cause the budget line to pivot inward.

The interaction between the budget line and indifference curves provides insights into consumer behavior, including how changes in income and prices affect consumption patterns and preferences. This graphical analysis is fundamental to understanding how consumers make choices in a world of limited resources and competing desires.

Q.6. What are the different Techniques of Demand Forecasting and how it will help in introducing the new product in the market

Ans. Demand forecasting involves predicting the future demand for a product or service. Accurate demand forecasting is crucial for businesses to plan production, inventory, marketing, and resource allocation effectively. There are several techniques of demand forecasting, each with its own strengths and weaknesses. When introducing a new product to the market, demand forecasting helps assess potential customer interest and estimate the level of demand the new product might experience. Here are some common demand forecasting techniques and how they can aid in introducing a

Qualitative Techniques: Qualitative methods rely on expert judgment, subjective data, and qualitative data to make predictions. These techniques are often used when quantitative data is unavailable or insufficient for new products. Qualitative

Techniques: Surveys, focus groups, and interviews can provide insights into consumer preferences, intentions, and perceptions regarding the new product.

Expert Opinion: Consulting industry experts or internal experts within the organization can provide valuable insights and informed judgments about the potential success of the new product.

Market Research: Qualitative techniques provide initial insights into consumer preferences and potential challenges. They help identify features and aspects of the product that might attract or deter customers.

Time Series Analysis: Time series analysis uses historical sales data to predict future demand based on patterns and trends. Common methods include moving averages, exponential smoothing, and ARIMA (AutoRegressive Integrated Moving Average) models.

How it helps: If the company has a similar existing product or a product with similar characteristics, historical sales data can be used to create forecasts for the new product's demand. However, if the new product is truly innovative, this method might not provide accurate predictions.

(c) Causal Models: Causal models consider factors that influence demand, such as economic indicators, competitor actions, advertising spending, and seasonality. Regression analysis is often used in this approach to identify relationships between demand and influencing factors.

How it helps: Causal models are useful when the introduction of the new product is affected by external variables, such as economic conditions or competitor strategies. They can help estimate how these variables might impact the demand for the new product.

(d) Simulation Models: Simulation models use computer-based simulations to forecast demand based on various scenarios. They can be useful when predicting demand for complex products or when considering multiple interacting variables.

How it helps: Simulation models allow businesses to test different assumptions and scenarios, helping them anticipate how the new product might perform under various conditions and inputs.

(e) Leading Indicators: Leading indicators are economic or market variables that change before the overall economy or market changes. By tracking these indicators, businesses can predict changes in demand.

How it helps: Leading indicators can provide insights into economic trends that might impact the launch of the new product, helping the company make informed decisions about timing and marketing strategies.

When introducing a new product, a combination of these techniques is often used to provide a comprehensive understanding of potential demand. The insights gained from demand forecasting assist in making informed decisions about production levels, marketing budgets, pricing strategies, and resource allocation, ensuring a smoother and more successful product launch. Keep in mind that demand forecasting is inherently uncertain, so it's important to monitor actual sales and adjust strategies accordingly as the new product enters the market.

Q.7. How can we differentiate the cardinal and ordinal Utility. Explain

Ans. Cardinal utility and ordinal utility are two different approaches to measuring and analysing consumer preferences and satisfaction in economics. They represent two distinct perspectives on how individuals make choices and rank their preferences for different goods and services.

1. Ordinal Utility: Ordinal utility is based on the idea that individuals can rank their preferences for various goods and services but without assigning specific numerical values to those preferences. In other words, ordinal utility focuses on the relative ranking of preferences rather than the intensity of preferences. It implies that individuals can tell which option they prefer more or less, but they cannot quantify by how much.

For example, if a person prefers Good A over Good B, we can say that Good A provides greater utility for them. However, ordinal utility doesn't tell us how much greater the preference is in quantitative terms.

2. Cardinal Utility: Cardinal utility involves assigning numerical values to preferences in a way that reflects the intensity of satisfaction or happiness a person

derives from consuming a certain quantity of a good or service. This approach suggests that preferences can be measured in a precise and quantifiable manner.

For instance, if a person assigns a utility value of 10 to consuming 2 units of Good A and a utility value of 5 to consuming 1 unit of Good B, cardinal utility would suggest that the person derives twice as much satisfaction from consuming Good A compared to consuming Good B.

Key Differences:

Measurement: The main difference between ordinal and cardinal utility lies in measurement. Ordinal utility involves ranking preferences without assigning specific numerical values, while cardinal utility assigns numerical values to preferences.

Quantifiability: Ordinal utility is not concerned with measuring the exact intensity of preferences; it's focused on the order of preference. Cardinal utility seeks to measure preferences in a quantitative manner.

Comparative vs. Absolute: Ordinal utility allows for comparing preferences (greater, lesser, or equal), while cardinal utility attempts to quantify the absolute level of satisfaction or happiness.

Subjectivity: Ordinal utility is less subjective because it only requires individuals to indicate their preferences in relative terms. Cardinal utility involves assigning numerical values, which can be more subjective and challenging to determine accurately.

Use in Economics: Ordinal utility is more commonly used in modern economic theory due to its simplicity and the fact that it doesn't rely on quantifying subjective experiences. Cardinal utility was used historically but has largely been replaced by ordinal utility due to its practical challenges and lack of empirical support.

In contemporary economics, ordinal utility is the foundation of consumer choice theory, where the focus is on the order of preferences and the concepts of indifference curves and budget constraints. Cardinal utility, while conceptually interesting, is less commonly used due to difficulties in assigning meaningful and universally applicable numerical values to subjective experiences of satisfaction.

Advantages and Disadvantages of Ordinal and Cardinal Utility

Ordinal Utility: Economies of scale and diseconomies of scale are concepts in economics that describe the effects of increasing or decreasing the scale of production on a firm's costs and efficiency. Here are the advantages and disadvantages of each:

Economies of Scale:

Advantages:

Cost Reduction: Economies of scale lead to lower average costs of production as the firm increases its output. This can result in increased profitability and a competitive advantage in the market.

Increased Efficiency: Larger production volumes often allow for better utilization of resources, specialized machinery, and more efficient production processes. This can lead to higher output per unit of input.

Better Bargaining Power: Larger firms often have more negotiating power with suppliers and can secure bulk discounts on raw materials and components, further reducing costs.

Spread Fixed Costs: Fixed costs (e.g., administrative expenses, rent) can be spread over a larger production output, leading to lower average fixed costs per unit.

Research and Development: Bigger firms can invest more in research and development, innovation, and technology, leading to improved product quality and diversity.

Disadvantages:

Complexity: As firms grow in size, managing operations becomes more complex, requiring more coordination and communication, which can lead to inefficiencies.

Bureaucracy: Larger organizations might face bureaucratic challenges that slow down decision-making processes and hinder flexibility.

Loss of Flexibility: Large firms might struggle to adapt quickly to changes in the market or shifts in consumer preferences due to their size and established processes.

Diseconomies of Scale Transition: There's a point beyond which further expansion might lead to diseconomies of scale, where costs start increasing due to complexities and inefficiencies.

Diseconomies of Scale:

Advantages:

Increased Flexibility: Smaller firms are often more flexible and able to respond quickly to changes in the market or customer demands.

Simplicity: Smaller firms generally have simpler structures, which can lead to more efficient communication and decision-making.

Personalization: Smaller firms can provide more personalized services and tailor products to specific customer needs.

Innovation: Smaller firms might be more innovative, as they can experiment and adapt more easily without being bound by established procedures.

Disadvantages:

Higher Costs: Smaller firms might face higher average costs of production due to less efficient resource utilization and inability to benefit from economies of scale.

Limited Bargaining Power: Smaller firms might have less negotiating power with suppliers, leading to higher input costs.

Risk of Closure: Smaller firms might face higher risk of failure due to lack of resources and economies of scale to withstand market challenges.

Limited Resources: Smaller firms might struggle to invest in research and development, technology, and marketing compared to larger competitors.

Therefore, the impact of economies and diseconomies of scale varies across market conditions, and the specific strategies of firms. The optimal scale for a firm depends on a careful balance between reaping the benefits of economies of scale and avoiding the pitfalls of diseconomies of scale.